

The Impact of Coal Mining on Drinking Water Quality, Monitoring, and Violations

Emil Kee-Tui and Nicolas Ramkumar

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Abstract

The Clean Air Act Amendments of 1990 placed limits on NO₂ and SO₂ emissions from coal fired power plants, significantly reducing acid rain. Power plants partly met the requirements by consuming low sulfur coal, coal with less than 2% sulfur by weight. The change in demand reallocated coal mining away from high sulfur coal mines to low sulfur coal mines. We exploit the time varying shift in demand for coal by sulfur content as an instrumental variable to measure the downstream impact of coal mining on drinking water quality provided by community water systems (CWS). An additional coal mine upstream of a CWS causes a large and statistically significant increase in CWS violations of the Safe Drinking Water Act (SDWA). To investigate the incentives to violate the SDWA, we measure the actual contaminants detected at CWSs since costs of SDWA compliance rise with pollution. We find that mean concentrations of inorganic chemical contaminants increase with upstream coal mining. To investigate the disincentives to violate the SDWA we measure the likelihood of CWSs facing formal versus informal enforcement action, formal action is a relatively higher costing enforcement action for being out of SDWA compliance compared to informal action. We find formal action is less likely as coal mining increases upstream. These results suggest that violations occur more frequently downstream of coal production because incentives

to violate increase and costs of enforcement action decrease with upstream coal mining.

1 Introduction

US coal mining in the near term is likely to continue in order to meet domestic and foreign demand for coal. Coal remains an important source of energy, providing 33% of global electricity in 2024 and 16% of US electricity in 2025 (International Energy Agency (2025)). Although coal has been rapidly losing market share from cheaper sources of energy such as natural gas and renewables such as solar and wind the second Trump administration has sought to slow down and reverse the decline in coal power generation (Trump, Donald J. (2025a), Trump, Donald J. (2025b), Trump, Donald J. (2025c); Daly (2025)). The environmental damages of coal will continue to be experienced. Coal produces a large amount of pollution at all stages of its life cycle, from mining to consumption and waste disposal. Coal mining creates acid mine drainage that leaches inorganic chemicals into surface and groundwater (Aragón and Rud (2016); Chu et al. (2023); Sigman (2002); Somerville, Mérel, and Hadacheck (2025); Lutz, Bernhardt, and Schlesinger (2013)). Coal ash waste from power plant's contains heavy metals that can enter the surface and groundwater (Austin (2020)). Coal combustion produces greenhouse gas emissions like CO₂, acid rain precursors like SO₂ and NO_x, and local air pollutants such as PM_{2.5}.

The impact of coal mining pollution on health in the US depends on the responses of community water systems (CWS) to upstream coal mining water pollution since 85% of the US population receives their drinking water from a CWS. CWSs regulated by the Safe Drinking Water Act (SDWA). CWSs are required to treat drinking water to meet legal standards set by state health and environmental regulations and federal regulations under the Safe Drinking Water Act (SDWA). The SDWA forms the regulatory floor for water quality at CWSs and relies on CWSs to self monitor and report their water quality to state regulators who are responsible for enforcing compliance. CWSs self monitor and

report water quality to regulators. CWSs violate the SDWA when they fail: To monitor their water and/or report tests to regulators on time (MR violation). Tested contaminant levels exceed the maximum contaminant level (MCL violation). MCL violations have greater penalties than MR violations. Enforcement attempts to be remedial rather than punitive. CWS incentives to meet SDWA depend on the cost of compliance and the penalty of non-compliance.

This paper asks if CWSs stay in SDWA compliance with upstream coal mining pollution? To answer this question we create a panel data set combining CWS characteristics, violations of the Safe Drinking Water Act with upstream coal production and characteristics. Violations are measured when pollution concentrations exceed the amounts allowed by law (maximum contaminant levels) and when CWS's avoid their responsibility to test and report water quality tests to administrators, resulting in a monitoring and reporting violation. The Acid Rain Program (ARP) of 1990 provides an instrument to measure the impact of coal mining on CWSs and the impact of upstream water pollution on CWS. The ARP's effect on coal production is a valid instrument because none of the provisions of the Acid Rain program or the production decisions of coal mines responding to the ARP were related to the characteristics of CWSs. An additional coal mine upstream of a CWS water intake leads to a decrease in CWS monitoring and reporting of arsenic and inorganic chemical contaminants. We do not precisely detect a change in the number of CWS violations exceeding the MCL for a contaminant concentration for an inorganic chemical. If MR violations hide MCL violations we cannot rule out that water is exceeding MCL levels.

The ARP was created by the Clean Air Act amendments (CAAA) of 1990 to reduce acid rain by limiting SO₂ and NO_x emissions from coal fired power plants. The ARP resulted in an increase in coal production in low sulfur coal beds, coal

with less than 2% sulfur content by weight, and a decrease in production in high sulfur coal beds since coal fired power plants are the largest consumers of coal in the US, and they primarily complied with the regulation by switching to low sulfur coal that produces less SO₂ when burned. After the ARP downstream, coal mining impacts were determined in part by the sulfur content by weight in coal beds upstream of CWS water intakes, the facilities that CWSs draw raw water to process for consumers. As coal production shifted to low sulfur coal beds, which areas increased their exposure and which areas reduced their exposure to coal mining water pollution? We answer this question by creating a panel of CWS water intake locations, the population relying on each CWS, annual coal production occurring upstream, and the amount of sulfur by weight in those upstream coal mines. The average number of customers relying on a CWS that was downstream of a coal mine fell by 82% between the periods 1985-1989 and 2000-2005, from an annual average of 1,285,00 to an annual average of 237,000.

US policy has attempted to address coal pollution by promoting “clean coal”. Clean coal refers to low sulfur coal, coal with less than 2% sulfur content by weight (Schweinfurth (2009)), or coal fired power plants that use SO₂ and NO_x scrubbers. “Clean” coal, in fact, is not clean and still emits greenhouse gases, particulate air pollution, and water pollution.

CWS incentives to meet SDWA depend on the cost of compliance and the penalty of non-compliance. We test whether violations increase because costs of compliance increase. To do so this paper asks whether upstream coal mining increases pollution concentration at CWS. To answer this question we create a panel data set of contaminant concentrations and cumulative upstream coal extracted since 1985. Every additional 10,000 tonnes of coal mined upstream of a CWS water intake leads to

CWSs would also increase MR violations (non-compliance) if the level of enforcement falls. This paper asks if coal mining raise enforcement at CWSs affected by coal mining? We create a panel of enforcement action and regulator behavior towards CWS and upstream coal mine and use the same instrument of the ARP after 1995. We find that formal enforcement is 5 percentage points lower for an additional upstream mine and there is more sanitary visits by 14 percentage points.

The literature on environmental self monitoring suggests that underreporting can occur if the benefits to the CWS of an MR violation versus an MCL violation exceed the enforcement penalty. The penalties for MR violations are lower than those for MCL violations creating an incentive for CWSs to undermonitor and report as a way to avoid an MCL violation. The regulator is more likely to respond to MCL violations with compliance agreements, penalties, criminal cases, and consent decrees. Whereas regulators are more likely to respond to MR violations by requiring the CWS to issue a public notification, remind the CWS to test or submit tests, and send notices to the CWS that a violation has occurred. Overall MR violations are penalized less harshly than MCL violations. When CWS's avoid their responsibility to test and report water quality when ambient pollution levels are rising this is an example of strategic violation behavior.

Regarding our ARP results, although we do not present health effects, the evidence on the effects of health from mining pollution suggests that the total health improvements are substantial (Parfitt (2024)). Aside from meeting its primary goal of reducing acid rain (Chestnut and Mills (2005); *Acid Rain Program Results* (2026)), the ARP improved PM2.5 levels and reduced mortality (Chestnut and Mills (2005); Chan et al. (2018); Alan I Barreca, Neidell, and Nicholas J Sanders (2021)), not without cost to power generators in installing

abatement technology (Chestnut and Mills (2005); Chan et al. (2018)), employment losses in high sulfur coal mining (*Impacts of the Acid Rain Program on Coal Industry Employment* (2001); Douglas and Wiggins (2015)), raising railroad transport prices (Busse and Keohane (2007)), and reducing crop yields by decreasing sulfates (Nicholas J. Sanders and Alan I. Barreca (2022)).

This paper Does coal mining raise water pollution at CWSs? and contaminant concentration levels

These results contribute to the literature on strategic environmental self reporting. This literature finds evidence that when a regulation applies to firms or activities that exceed a size or production threshold and firms know those thresholds they strategically maintain their size or production below that level to avoid environmental reporting requirements Benneer (2008). CWSs have been found to use testing strategically to avoid more serious MCL violations, Benneer, Jessoe, and Olmstead (2009) find that to avoid a total coliform rule that is calculated based on the share of positive total coliform samples out of a mandatory minimum samples systems most likely to get MCL for total coliform perform additional voluntary oversampling which increases the denominator in the calculation of the share of positive samples and reduces the likelihood of violations despite no change in water quality. Despite national regulation of water quality and most households accessing treated water US drinking water quality still has deficiencies that lead to significant costs and health effects. Based on bottle water sales as avoidance behavior for drinking water violations, Zivin, Neidell, and Schlenker (2011) estimates the cost of avoidance in 2005 at \$60 million. A large number of people in the US are affected by water quality violations at CWS, between 1984 and 2018 Allaire, Wu, and Lall (2018) estimate 9-34 million people on average are affected by violations for a contaminant that exceeds safe health levels. Particular water quality crises such as flint cost \$520

million in housing value alone Christensen, David A. Keiser, and Lade (2023). Manipulation can be countered by increasing scrutiny of CWSs such as Andarge et al. (2025) who find that manipulation of reporting under the lead and copper rule was prevalent before 2016 but not after 2016 when the flint, MI crisis increased scrutiny of CWSs. [THESE RESULTS CONTRIBUTE TO TWO STRANDS OF LITERATURE: US WATER QUALITY, CWS STRATEGIC VIOLATIONS, and ENVIRONMENTAL SELF REGULATION] Water quality in the US (David A Keiser and Shapiro (2019) find that between 1972 and 2001, the share of waters safe for fishing grew by 12 percentage points; David A Keiser (2019) This paper’s primary research design uses upstream concentrations of phosphorus as an instrumental variable for local county-level pollution concentrations. The validity of this approach rests on the assumption that pollution from far away distances affects recreation, but only through its effect on local pollution concentrations. **keiser2023us** find drinking water readings exceeding current health standards fell by half between 2003 and 2019. Loans to public drinking water systems through the Safe Drinking Water Act contribute to the decline in water pollution. SDWA loans reduce mortality among older adults). Studies of CWS’s have found evidence of strategic violation before but have relied on violation data without ambient water pollution readings. This study provides stronger evidence of strategic violations because our data combines ambient pollution levels and violations and we show concurrent rising monitoring and reporting violations and ambient pollution levels. Using coal mining as the pollution source gives two advantages, the ARP had nothing to do with local water quality and was a national regulation on air quality that changed the location of coal mining and which CWS’s were downstream of coal mining. The changes in coal production resulting from the ARP is therefore unrelated to how well a CWS treats its water or has a tendency to engage in strategic

violation and changes in coal production changed and provides an instrumental variable to predict a causal effect of coal mining on drinking water quality. This instrument is similar to that used by Douglas and Wiggins (2015). [HOW OUR FINDINGS FIT INTO THE LITERATURE ON ENVIRONMENTAL MONITORING AND CWS STRATEGIC VIOLATIONS][Strategic self-monitoring of pollution, strategic pollution sampling, and regulator action (Helland (1998); Stafford (2002); Benneer, Jessoe, and Olmstead (2009); Duflo et al. (2013); Duflo et al. (2018); Mu, Rubin, and Zou (2024))]

CWSs are different from the traditional private polluting firms that generate the pollution and are responsible for reporting to the regulator. In that context firms have incentives to not truthfully report their true emissions and even third party auditors have an incentive to underreport to get return business from the polluting firm (Duflo et al. (2013)). Reporting can also be used as a way for a regulated firm to signal cooperation with a regulator to signal willingness to cooperate with regulator and move to less costly monitoring by the regulator (Helland (1998)). We see evidence of CWSs not reporting rather than reporting false reports. CWSs are also not the firms creating the pollution but are responsible for removing upstream pollution. Mu, Rubin, and Zou (2024) develop a method to detect local governments strategic non-monitoring of air quality reading equipment when they anticipate air quality alerts or pollution spikes. Their method is an event study of missing monitoring data on the day when air quality alerts affect their county. Our findings are consistent with literature on CWS violations of drinking water quality that suggest CWS's strategically avoid the more serious violations that come with detecting a higher concentration of contaminants by not reporting or testing their water (Wallsten and Kosec (2008) suggestive but inconclusive that MR is used to mask more serious MCL violations; Foster et al. (2019) does not find an increase in MR when arsenic MCL was lowered - they find

that reductions in MCL were true reductions and not associated with masking with MR. *EPA Claims to Meet Drinking Water Goals Despite Persistent Data Quality Shortcomings* (2004) finds that between 1996-2001 MR violations were underreported to SDWIS with ranges from 9-23%. This is possible because CWS water quality is self monitored and regulators rely on the results of their submitted records to issue violations for exceeding the allowable limit. Empirical evidence indicates that formal penalties used to enforce pulp, paper, and paperboard mill plants to comply with the Clean Water Act generated substantial deterrence for non-penalized plants. Shimshack and Ward (2005) find that a single enforcement fine reduces the statewide violation rate by roughly two-thirds in the following year, primarily through spillover effects to uninspected systems.

We create a panel data set combining CWS characteristics, violations of the Safe Drinking Water Act, and contaminant concentration levels with upstream coal production and characteristics. Coal production the EPA datasets on CWSs including, SDWIS which contains CWS characteristics such as population served, groundwater or surface water, public or private ownership etc, We also use data on the location and type of CWS water intakes. The impact of coal mining on CWSs is measured with CWS violations of the SDWA from SDWIS, it includes our main source of CWS and CWS regulator behavior data including details on CWS violations of the Safe Drinking Water Act, this is the national CWS drinking water regulation, and enforcement actions regulators made at CWS. And actual measurements of some chemical contaminants from the second and third rounds of the 6 year review process. The location and production at individual coal mines is measured used Annual coal production at individual coal mines from the Energy Information Administration and the location of those mines from the Mine Safety and Health Administration. The

coal quality at those mines is estimated using USGS drill sample data. A national USGS watershed connectivity dataset is used to associate CWS water intakes with coal production in upstream watersheds.

A Community Water System (CWS) provides piped drinking water for human consumption to at least 25 year-round residents through at least 15 service connections.

For a majority of US households, 85%, household water is obtained through one of 144,000 community water system's (CWS) that supply other households, may be publicly or privately owned, and are mandated by local, state, and federal regulation to treat the water. CWSs enable public health and environmental efforts to protect people from water pollution be feasible at the final consumption stage unlike air pollution which is harder to contain, filter, and clean at the point where it is breathed. CWSs are regulated to supply and treat water at least adhering to minimum levels of quality through the federal Safe Drinking Water Act of 1974. The regulation is intended to ensure that at any of the 144,000 CWSs around the US the quality of water exiting a person's pipe or faucet meets a minimum standard. The source water that enters the systems is not uniform across the US and requires each CWS to employ unique and inventive solutions to bring their local source water up to the national standard of testing and quality. The rules require testing and CWSs may have local knowledge over quality, preferences over quality, and costs. CWSs internalize the costs of the water quality standards but do not always internalize the benefits. To internalize the cost the regulators employ penalty schedules that Some CWSs may find it in their best interest to do less than required and some regulators may find it in their best at each CWS requires addressing the unique local water quality conditions because The waters of the US The intake can be interceded at the point of consumption A portion of the risks to human health from water

pollution can in theory be abated by strict enough regulation and adherence to that regulation at CWSs. CWSs are regulated and places to intervene to improve water pollution exposure but they respond to costs of cleaning and may not do what the law demands of them

We study the impact of coal mining on drinking water quality violations under the Safe Drinking Water Act (SDWA) at U.S. Community Water Systems (CWSs). CWSs provide over 85% of people in the US with drinking water; understanding their response to pollution is vital to protect human health. We use an instrumental variable to identify variation in coal production that is exogenous to its impact on drinking water quality. The instrument uses the change in demand for coal with different amounts of sulfur that resulted after the implementation of stage I of the Acid Rain Program in 1995.

A third of global electricity is generated by coal (International Energy Agency (2025)). The Trump administration has prioritized maintaining and expanding coal power generation and mining Trump signed several pro-coal executive orders: Trump, Donald J. (2025a), Trump, Donald J. (2025b), Trump, Donald J. (2025c) And opened millions of acres of federal land for coal mining (Daly (2025)).

The negative environmental impacts of coal are well studied and documented: Air quality. Climate warming greenhouse gas emissions. The environmental impact of coal mining has also been documented (Muller, Mendelsohn, and Nordhaus (2011)). Coal’s impact on drinking water quality has been less well studied.

Coal’s impact on the environment Main water impacts: Inorganic chemical pollution and pH decreases.

Polluted drinking water is the primary pathway through which water pollution from coal mining threatens human health. CWS’s provide drinking water to 85% of US households. Drinking water quality is regulated by the Safe Drinking

Water Act (SDWA). It is not known whether CWS water treatment is enough to nullify coal mining pollution.

What impact does coal mining have on Community Water Systems' (CWS) Safe Drinking Water Act (SDWA) violations? Empirical approach: Data on CWS violations, water intakes, and coal quality and production upstream of intakes from 1985 to 2005. IV for coal mines upstream of CWS using the differential impact of the Acid Rain Program on coal production based on sulfur percentage. What we find: Increases in drinking water quality violations. Driven by violations from failing to conduct and submit water quality tests to regulators on time. Violations from exceeding contaminant levels are economically and statistically insignificant. This pattern provides suggestive evidence that community water systems engaged in strategic violation substitution. Dangerous for health since MCL violations trigger protective avoidance behavior (Zivin, Neidell, and Schlenker (2011)).

1. Enforcement action in other contexts has shown to depend on other characteristics of the polluter in addition to the pollution. To look at just the enforcement action ignores the potential selection bias (Hanna, Oliva, et al. (2010) finds selection bias in clean air act). We can measure impact of enforcement action or violation after convincing causal identification of the violation.

2. I am interested in estimating the regulator's enforcement response to violations. But enforcement will be autocorrelated with previous enforcement. A system's violations and the regulators enforcement action will both be affected by past enforcement and violations and the enforcement and violation of other systems.

Contribution, Strategic self-monitoring of pollution, strategic pollution sampling, and regulator action (Helland (1998); Stafford (2002); Benneer, Jessoe, and Olmstead (2009); Duflo et al. (2013); Duflo et al. (2018); Mu, Rubin, and

Zou (2024)).

Impacts of the Acid Rain Program (Schmalensee and Stavins (2013); Chan et al. (2018); Busse and Keohane (2007); Li and Chen (2024); Alan I Barreca, Neidell, and Nicholas J Sanders (2021))

Pollution impacts of resource extraction and upstream sources (Aragón and Rud (2016); Chu et al. (2023); Sigman (2002); Somerville, Mérel, and Hadacheck (2025); Lutz, Bernhardt, and Schlesinger (2013); Hill and Ma (2017))

This follows the standard approach in the literature measuring the impact of policies or pollutants on water quality (**keiser; keiser;china cancer upstream**).

Lit review: this paper contributes to lit on impact of land use on water quality Somerville, Mérel, and Hadacheck (2025)

impact of coal supply chain (social cost of carbon, health and air pollution)
impact of mining on downstream health (Parfitt (2024))

PWS violations and regulator behavior (?)

Water pollution levels in drinking water and non-drinking water have fallen in the US since the passage of the Clean Water Act of 1972 and the Safe Drinking Water Act of 1974 (David A Keiser and Shapiro (2019); **keiser2023us**). Despite the overall progress, millions of people in the US struggle to access safe drinking water either because they are served by drinking water systems that violate health standards or because they face water affordability barriers (Allaire, Wu, and Lall (2018); **cardoso2022water**). Community water system's (CWS) are the main source of drinking water for 85% of US households.

The degree of local, state, and federal involvement varies from complete local control to authorities sharing governance for infrastructure financed through the Water Infrastructure Finance and Innovation Act (WIFIA) and the Clean Water State Revolving Fund (CWSRF), to total federal control, such as for a federally owned CWS (Mullin and Daley (2018)).

CWS's can take pre-emptive actions to protect the water at the source before intake into the system. New York City has protected the reservoirs in the adirondaks that it sources its drinking water from by purchasing the land surrounding the reservoirs and heavily restricting the activities that can take place on the land. CWS's treat the water after intake to bring it to potable standards that meet the SDWA requirements. Removing contaminants requires infrastructure and water treatment inputs. Despite the efforts CWSs make to treat their water, pollution in source water can result in CWS water quality falling below legal standards resulting in a violation. Heavily polluted source water and fluctuations in the quality of the water make treatment difficult and violations at CWSs more likely.

2 Background

2.1 The Acid Rain Program (ARP)

The Acid Rain Program (ARP) was signed into law on November 15, 1990 as amendments to the Clean Air Act to reduce sulfur dioxide (SO₂) and nitrogen oxides (NO_x) emissions, the primary precursors of acid rain, from fossil fuel burning power plants. Total annual US SO₂ and NO_x emissions goals were set at 10 million and 2 million fewer tons below their 1980 emission levels. The emissions reductions were met by creating emissions cap and trading markets for polluters, installing scrubbers on power plants, and power generators fuel switching. The ARP achieved 100% compliance (Lattanzio (2022)).

In two phases the SO₂ emissions from facilities built before November 15, 1990 were capped at a fraction of each facility's individual baseline SO₂ emissions in 1985. In Phase I, the EPA and states with permitting capacity issued emission permits and allowances to 110 high polluting, mostly coal, electricity

generating facilities. Facilities targeted under phase I were required to achieve compliance with SO₂ emission reductions by 1995. Facilities built after November 15, 1990 were not issued any SO₂ allowances and any emissions they produced had to be purchased from facilities with allowances through an auction. Facilities were required to install SO₂ and NO_x monitors and were fined when their emissions exceeded their allotted emission limits or the sum of emission limits they purchased on the SO₂ and NO_x trading markets from other emitters. Phase II capped the SO₂ emissions from facilities with power generation capacity greater than 75 megawatts. The deadline for compliance with phase II was January 1, 2000. We focus on the facilities affected by phase I because they are the largest power plants and coal consumers.

As opposed to SO₂ emissions, the ARP did not limit total NO_x emissions from sources; instead it limited the NO_x emissions per mMBTU of fossil fuels burned. NO_x limits affected coal fired facilities at the same time they came under SO₂ emission limits. Phase I NO_x rules were promulgated in 1995. Phase II NO_x regulation was promulgated in 2000.

Fuel switching to low sulfur coal was a primary method by which electricity generators met the ARP requirements. Since the electric power industry accounted for 90% of coal demand in 2000 (Freme (2000)), the ARP increased the demand for low sulfur coal and reduced the demand for high sulfur coal (*Impacts of the Acid Rain Program on Coal Industry Employment* (2001); Douglas and Wiggins (2015)).

2.2 Coal mining water pollutants

Coal mining releases heavy metals, radioactive elements, non-organic compound pollutants, and polycyclic aromatic hydrocarbons (PAHs) into the environment (Rouhani, J. Skousen, and Tack (2023); Boente, Baragaño, and Gallego (2020);

Liang et al. (2017)). Acid mine drainage is a common procedure that produces the inorganic chemical pollutants that are dissolved into water and transported into groundwater or surface water. Acid mine drainage occurs when rocks containing sulfides are exposed to air and water to begin the process of oxidizing and decomposing (Ceto and Mahmud (2000)). The oxidized minerals form sulfuric acid, metal ions and sulfates which can be carried into the surface or groundwater from water runoff (Akcil and Koldas (2006)). Group II and the transition metals such as Iron (Fe) and aluminum (Al), dominate the dissolved cations in AMD while sulfate and bicarbonate are the most common anions (J. G. Skousen, Ziemkiewicz, and McDonald (2019)). Acid mine drainage is more common when the minerals contain high levels of sulfur, making acid mine drainage increase in incidence with coal sulfur level (Schweinfurth (2009)).

Coal washing involves activities that reduce impurities in coal before use for example using water or chemical solutions to remove impurities. Washing with water creates a slurry that contains the removed impurities such as arsenic, lead, chromium, beryllium or nickel which contaminate surface or groundwater (Duhigg (2009)). Coal mines can process their coal to remove some sulfur in order to meet the demand for low sulfur coal but this can also be done by coal users along the production chain.

2.3 Community Water Systems (CWS) and the Safe Drinking Water Act (SDWA)

Community water systems (CWS) are entities which provide water to at least 25 consumers through at least 15 connections throughout the year. They can be private or publicly owned, source their water from ground water, surface water, or another CWS, and serve small rural communities up to the largest metropolitan areas. Under the Safe Drinking Water Act (SDWA), CWSs are

responsible to provide water to consumers that meets quality standards set by the USEPA. State agencies are responsible for enforcing CWS compliance with the SDWA. The USEPA oversees state enforcement of the SDWA and promulgates contaminant lists that CWSs must monitor for and remove from their water up to a safe level unique for each contaminant. The USEPA prescribes a maximum contaminant level (MCL) for each CWS which is the maximum amount of that pollutant that can be detected without being in violation of the SDWA. Between 1985 and 2005, the EPA promulgated 20 rule changes that expanded the list of regulated contaminants, changed how and how often CWSs monitor and report tests, and added treatment techniques CWSs must apply to their water treatment.

Water is regulated in the SDWA on a contaminant by contaminant basis and can be grouped into inorganic chemicals, volatile organic chemicals, synthetic organic chemicals, radionuclides, and bacteria. The SDWA MCL requirements for arsenic and nitrates remained the same between 1985 and 2005.¹ Once VOC and SOC MCL's were established their MCL's did not change between 1985 and 2005. A subset of SOCs were regulated contaminants across the time period from 1985 to 2005.² VOC MCL's contaminants were added to regulation in 1987.³ Total coliform MCL's were regulated from 1976 but were updated to be stricter in 1989, a total coliform MCL was no more than % 5 of monthly

¹The other inorganic chemicals regulated between 1985 and 2005 began with barium, cadmium, chromium, lead, mercury, selenium, and silver, with the addition of antimony, asbestos, Beryllium, cyanide, nickel, and nitrite by 2005.

²In 1985 SOC MCL's were established for contaminants 2,4-D, 2,4,5-TP (Silvex), endrin, lindane, methoxychlor, and toxaphene. By 2005 the list included acrylamide, adipate, di(2-ethylhexyl), alachlor, aldicarb, aldicarb sulfone, aldicarb sulfoxide, atrazine, carbofuran, chlor-dane, dalapon, dibromochloropropane (DBCP), dinoseb, dioxin (2,3,7,8-TCDD), diquat, endothall, epichlorohydrin, glyphosate, heptachlor, heptachlor epoxide, hexachlorobenzene, hexachlorocyclopentadiene, oxamyl (Vydate), PAHs (Benzo(a)pyrene), PCBs (Polychlorinated biphenyls), pentachlorophenol, phthalate, di(2-ethylhexyl), picloram, and piazine.

³In 1985 VOC contaminants were benzene, carbon tetrachloride, 1,2-dichloroethane, p-dichlorobenzene, 1,1-dichloroethylene, 1,1,1-trichloroethane, trichloroethylene, vinyl chloride, by 2005 the list of VOC MCL's added (mono) chlorobenzene, dichloromethane, cis-1,2-Dichloroethylene, trans-1,2-dichloroethylene, 1,2-dichloropropane, ethylbenzene, ethylene dibromide, o-Dichlorobenzene, p-Dichlorobenzene, styrene, tetrachloroethylene, toluene, 1,1,2-trichloroethane, 1,2,4-trichlorobenzene, trichloroethylene, vinyl chloride, and xylenes.

samples could be positive.

The SDWA relies on CWSs self-monitoring and reporting their tests to regulators. CWSs are issued violations by the regulator when they failure to comply with the SDWA. The regulator issues an MCL violation when a CWS detects and reports a contaminant within its water at a level that exceeds its MCL. The regulator issues a monitoring and reporting (MR) violation when a CWS fail to correctly test, monitor, and submit water test results to regulators. Many contaminants required their own tests had their own methods required by the EPA which may be done on site or required a laboratory. The CWS had 40 days to report the results of a test to the state regulator and 48 hours to report a failure to meet the testing or monitoring standard. A CWS that does not monitor is not presumed in violation of the MCL, only of the monitoring requirement itself.

Monitoring and reporting requirements were found to be difficult to meet. A survey of Indiana CWS managers who responded to a 1980 survey on the impact of the SDWA said that the SDWA's reporting requirements had the greatest impact on their systems, while MCL had the least (Oleckno (1982)). The MR testing schedule was made more complicated when it underwent changes. In 1991, CWSs were required to conduct tests for four consecutive quarters to establish baseline levels for inorganic chemicals (IOCs), VOCs, and SOCs if they could not show there was no risk of contamination and had not tested after 1990. Afterwards, groundwater systems were required to test every three years and surface water systems were required to test annually (56 FR 3526, 1991). The first three-year compliance period ran from 1993-1995, the second ran from 1996-1998, and the third compliance period (1999-2001). The next nine year timeline began in 2002 with the first cycle ending in 2004 and the next cycle starting in 2005.

2.4 Regulator Enforcement and Compliance

States are the primary enforcers of the SDWA and received primacy once they developed the administrative capacity and state laws to enforce and monitor CWS compliance with the SDWA. The enforcement sequence for an MCL violation begins a CWS self reporting or state inspection finding a contaminant exceeding an MCL. The regulator then employs informal compliance assistance (letters, notices of violation, compliance plans) and, if violations persist, formal administrative actions. The EPA will issue a federal administrative compliance order or civil action if the state has not acted within 30 days. The SDWA forms the floor of water quality and states are able to pass stricter drinking water quality regulations. Primacy states receive and review water quality contaminant tests from CWSs and are responsible for enforcing CWS monitoring schedules, issuing violations for SDWA non-compliance, bringing violating CWSs into SDWA compliance, and issuing exemptions and variances for some rules to CWSs.

Except for a subset of violating circumstances and mandatory notifications to consumers that CWSs must provide when in violation, the SDWA gives primacy states wide discretionary power to tailor compliance plans to each CWS and violation. The SDWA gave states the legal authority to conduct sanitary visits and investigations of any CWS, but otherwise, states were empowered to devise their own enforcement techniques and create specific compliance plans as long as those plans were enacted quickly enough and brought the CWS back into compliance. Enforcement usually took the form of compliance plans requiring the CWS to directly address the cause of the violation and rarely included civil or criminal financial penalties. Compliance plans could still be costly to the CWS if they required installing treatment infrastructure or conducting tests.

Without a prescriptive enforcement legal requirement for noncompliance, in

theory, each state could design a tailored compliance plan for a particular violation at a particular CWS. Each pollutant may require a different technology, and there may exist multiple technologies that have different costs or technical requirements that a CWS may or may not be able to maintain or pay for depending on the resources and staff at the CWS.

Civil action in the courts brought by citizen suits or the USEPA when the state lacked primacy or failed in its duty to enforce the SDWA was another direct tool of SDWA enforcement. Civil actions in courts could force any government agency or CWS to undertake action to comply with the SDWA or face civil penalty. Between 1974 and 1995, fines were capped at \$5,000 per violation per day; the 1996 Amendments raised this to \$25,000 per day (adjusted for inflation in subsequent years). USEPA referred on average approximately 420 cases per year to DOJ for civil or criminal judicial enforcement between 1998 and 2011; total civil judicial penalties averaged approximately \$93 million per year (Shimshack 2014, p. 344). Ordinary SDWA noncompliance was not subject to criminal liability; criminal provisions are reserved for tampering with a water system (up to 20 years imprisonment, 42 USC §300i-1) and willful underground injection control violations.

The SDWA also enforced compliance by requiring CWSs to notify consumers of any violation, variance, or exemption they had received. variances and exemptions allowed a CWS to not comply with an MCL or MR requirement if the state determined that there was limited health effects of allowing the variance, and the risk of the contaminant was low after an initial monitoring period and the hydrogeology of the area supplying water does not contain sources of that pollutant. After the 1996 SDWA amendments all CWS had to send consumer confidence reports to their customers. These reputational compliance tools have been shown to be effective at inducing compliance at CWSs (Benneer and Olm-

stead (2008)).

3 Theoretical Framework

We adapt a static model of a regulator and a CWS from Kang and Silveira (2021), which is an adaptation of the one developed by Mookherjee and Png (1994), to provide theoretical context and predictions about CWS behavior under SDWA regulation after an increase in coal mining upstream of CWS water intakes. CWSs choose a negligence level $a \in [0, \bar{a}]$ which measures how much they avoid the requirements of drinking water quality regulations, from maximum negligence $\bar{a}$ to full compliance 0. Failing to conduct or submit tests to the regulator is an example of CWS negligence. Regulators do not directly observe CWSs choice of a . CWSs face a compliance cost, θ , that is a random variable drawn from Θ which is a strictly increasing and continuously differentiable function $F(\cdot)$ with support $(0, \bar{\theta})$. CWSs know their draw of θ , but regulators only know the distribution of Θ . A compliance cost is determined by both a CWS's ability to provide drinking and manage a system, for example the skills of managerial and technical staff, and by the costs the CWS spends to treat the water, for example the cost of filtration equipment or sterilization chemicals. assume compliance costs depend explicitly on the amount of upstream coal mining, such that $\theta(m)$ is continuously differentiable and non-decreasing in m .

CWSs receive a benefit $\theta b(a)$ from having compliance costs θ and choosing a negligence level a . A CWS's benefit from not conducting a test for a specific contaminant are the labor and lab fees for sampling they did not have to spend. The opportunity cost to the CWS of choosing a compliance level a relative to maximum negligence $\bar{a}$ is $\theta[b(\bar{a}) - b(a)]$.

The number of violations a CWS incurs is a random variable K that follows a Poisson distribution with mean a . Given negligence level a , observed violations

follow a Poisson process:

$$Pr(K = k|a) = e^{-a} \frac{a^k}{k!}$$

The penalty schedule the CWS faces for incurring k violations is given by $\epsilon(k)$ which is non-decreasing and continuously differentiable. The penalty schedule is a function that includes the likelihood of the regulator enforcing the penalty at each violation and the cost of the violation. The regulator can only base the penalty on k because it cannot observe negligence or cost of compliance. Assuming the CWS is risk neutral the expected penalty for a risk neutral CWS is

$$e(a) \equiv \exp(-a) \sum_{k=0}^{\infty} \frac{\epsilon(k)}{k!} a^k. \quad (1)$$

The expected payoff to the CWS from negligence level a is

$$-\theta(m)[b(\bar{a}) - b(a)] - e(a). \quad (2)$$

A negligence schedule, $a(\cdot)$, is implemented by a penalty schedule, $e(\cdot)$, if $a(\theta)$ maximizes equation 2 for all $\theta \in \Theta$. Combining the first order condition with the $a(\cdot)$ implemented by $e(\cdot)$ gives

$$\theta b'[a(\theta)] = e'[a(\theta)] \quad (3)$$

whenever $a(\theta) > 0$.

Proposition 1: CWSs choose higher negligence levels as compliance costs increase, $\partial a / \partial \theta > 0$.

Assuming $b(a)$ is a strictly increasing function and the F.O.C. in equation 3

solves the CWS maximization,

$$\frac{\partial a}{\partial \theta} = -\frac{b'(a)}{\theta b''(a) - e''(a)} > 0$$

CWS's with higher compliance costs, for example, those with more expensive to run filtration machinery, find the marginal benefit of an additional unit of negligence $\theta b'(a)$ larger relative to the marginal expected penalty $e'(a)$. They optimally reduce testing and compliance and increase the number of monitoring and reporting violations.

Proposition 2: More upstream mining raises negligence, $\partial a / \partial m > 0$.

$$\frac{\partial a}{\partial m} = -\frac{\theta'(m)b'(a)}{\theta(m)b''(a) - e''(a)} > 0$$

When mining raises pollution levels which increases the costs of water treatment and the compliance costs, the optimal negligence level increases, reducing testing and reporting and increasing the number of monitoring and reporting violations.

The increase in negligence and monitoring and reporting violations following an increase in coal mining can be offset by regulator by increasing enforcement of CWS violations.

Proposition 3: Lower penalties for violations reduces compliance and increases the number of monitoring and reporting violations. Assume higher penalties at every k increase the penalty schedule $\epsilon(k)$ by a fixed amount $\lambda > 1$, such that $\lambda \epsilon(k)$, then $\partial a / \partial \lambda < 0$.

$$\frac{\partial a}{\partial \lambda} = \frac{e'(a)}{\theta b''(a) - \lambda e''(a)} < 0$$

4 Data

4.1 River and stream connectivity

In order to measure which CWSs are downstream of coal mining, we measure the hydrological connectivity of mining and CWS water intakes using the USGS NHDplus watershed connectivity data. The NHDplus data identifies geographical boundaries where rainfall would drain into the same stream segment. Watershed's are assigned a unique 2-12 digit code known as a HUC (hydrological unit code). The number of digits corresponds to the size of the area, with the geographical area decreasing with the number of digits; for example, the lower 48 US states are divided into 18 HUC02's and over 87000 HUC12's (see figure 1). HUC12 watersheds are the smallest mapped watersheds available for the entire continental US and are small enough that coal mining activities may impact the water in the HUC12 directly downstream. However, HUC12's are also large enough that activity that takes place at the outlet of a HUC12 may not affect activity at an upstream inlet within the same HUC12.

CWS water intake locations are provided at the HUC12 level by the EPA. It is possible for CWS's to obtain water from multiple HUC12's. We consider a CWS to be downstream of a coal mine if any of its water intakes are located in a HUC12 directly downstream of a coal mine and none of its intakes are located in a HUC12 where mining took place during the sample period.

4.2 Coal Quality and Coal Mining Data

Individual coal mine production data was obtained from the Energy Information Association (EIA). It includes tonnes of coal produced by each registered mine in each year from 1985 to 2005. EIA coal mines are matched to latitude and longitude coordinates from the Mine Safety and Health Administration (MSHA).



Figure 1:

Figure 2 shows the HUC12's where mine production data from EIA was able to be matched to MSHA data of mine coordinates.

In order to match coal production to the geographic scale of CWS drinking water intakes we aggregate individual coal production to the HUC12 level by taking the sum of active mines in the watershed or the sum of coal mined in the watershed. The coal production data includes all coal mines that are one HUC12 directly upstream of a HUC12 where a community water system's water intake is located.

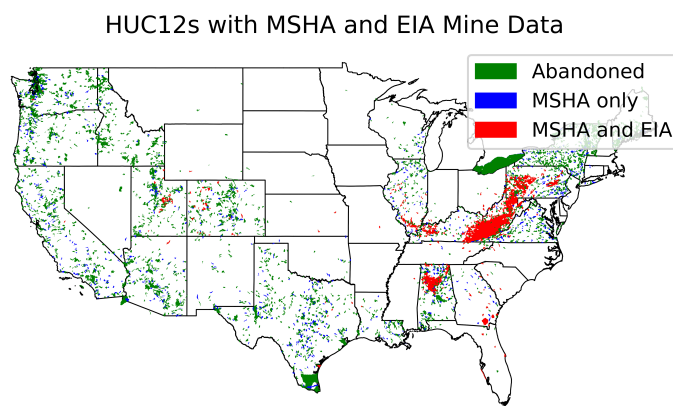


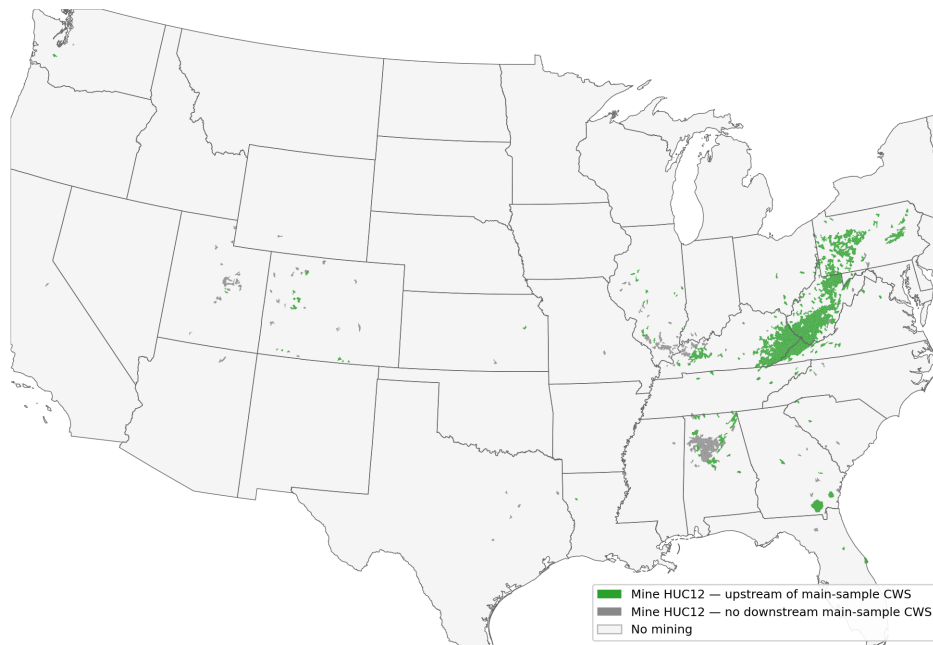
Figure 2:

The weight of sulfur by weight present in coal is obtained from the USGS COALQUAL dataset. The USGS COALQUAL dataset is a sample of boreholes drilled into coal beds across the US and sampled for its characteristics including sulfur by weight. Each sample includes the latitude and longitude coordinates which we match to the HUC12 level. We calculate the weight of sulfur as a percent of coal's weight in the HUC12 watershed as the mean coal sulfur percentage of all COALQUAL samples located within 30 km of the HUC12 watershed. This procedure is important for establishing coal quality that can be measured at the same geographical level as the CWS and coal mining data. It also approximates coal quality into watersheds that may not have coal production during the sample time period. It is reasonable to approximate that in a relatively small area, such as neighboring HUC12's, that coal may be present in

the ground even if its unexploited. Upstream coal sulfur could be mismeasured if it has extreme variation within a HUC12. Some "high sulfur" HUC12s may actually host low-sulfur seams. HUC12's are geographically small areas that on average have low variation in measured coal sulfur levels for samples that are used to estimate coal HUC12 coal sulfur levels.

Figure 5 plots a histogram of the sulfur percentage of the coal bed within the HUC12's where a CWS water intake and coal production are co-located. The sulfur content of HUC12 coal beds ranges from zero to 4.5%, with most of the HUC12 watersheds below 1.5%. Slightly more than half of HUC12's are low sulfur coal beds.

HUC12's are only included in our dataset if we are able to match COALQUAL samples. Map 4.2 shows the location of HUC12's in the sample that have CWS intakes and COALQUAL samples.



4.3 CWS characteristic, regulator, and violation data

We form an annual panel of all CWSs that were active from 1985 until 2005 and had an intake in a HUC12 that was directly downstream from an active coal mine during at least one of the years between 1985 and 2005, and had no water intakes in a HUC12 with a coal mine that was active during any time between 1985 and 2005. This sample contains 340 unique CWSs.

We use the EPA’s Safe Drinking Water Information System (SDWIS) Enforcement and Compliance History Online (ECHO) data to construct control variables of the number of CWS facilities, outcome variables of SDWA violations, outcome variables of regulator visits, and outcome variables of regulator SDWA enforcement action.

We control for the date of construction and decommission of CWS water intakes. This is the only time varying CWS characteristic we use. SDWIS contains only a snapshot of other system characteristics as they appeared during the most recent SDWIS update and does not vary with time during the period of our study. Therefore we cannot control for time varying CWS characteristics such as size of the community served and ownership type.

The SDWIS violation data uniquely identifies each CWS violation of the SDWA and includes the start and end date of the violation, the type of SDWA rule violated, the contaminant, the severity, the agency responsible for enforcing the violation, and the penalty or compliance action the agency took to bring the CWS back into compliance. We create a violation outcome variable that measures the share of the year a CWS exceeded the maximum contaminant level (MCL) or failed to meet the monitoring and reporting (MR) requirement for an arsenic, nitrate, and inorganic chemical rule. We limit our analysis to SDWA contaminant violations for arsenic, nitrates, and a general separate category of IOC due to their prevalence in the data, clear association with coal mining water

contamination, and stable MCL's over the time frame.

The share of the year spent in violation does not necessarily reflect that the CWS was violating continuously during that period. Figure 4.3 plots histograms of the length of arsenic, IOC, radionuclide, and nitrate violations. There is a tendency for length of time of violations to line up multiples of year quarters such a year, half a year, a quarter of a year, or three quarters of a year. This is evidence that violations are often triggered when CWS fail to meet quarterly testing schedules and then last a set period of time when they are issued. Nonetheless we measure violations using length of time out of the year in violation to exploit variation and information that the length of violations contains such as severity with more serious violations lasting longer.

Table 1 presents means and standard deviations of the number of days out of a year a CWS spends with an IOC MR and MCL violation. On average a CWS spends between 10 to 15 days out of a year with an MR violation and less than a day with an MCL violation. This reflects MCL violations are health dangers and are resolved as fast as possible in order and the rarity of MCL violations where in a given year most CWSs do not receive an MCL violation. The large standard deviation of 56 to 67 days reflects the multi-modal distribution that characterizes the distribution of the number of days a CWS is in violation. Many CWSs do not receive an MR violation in a year and those that do tend to experience it for a multiple of a quarter of a year.

Table 2 presents summary data of the MR and MCL violations that occurred at the 340 CWSs in the main sample between 1985 and 2005. There are only 15 MCL violations relating to arsenic, nitrate, and IOC. This makes MCL violations very rare in the data. Little can be extrapolated from 15 observations however most of MCL's were issued after a single contaminant sample exceeded the threshold and the rest were after the average of a group of samples exceeded

the threshold.

MR violations are more common, there are 1,233 MR violations. Nitrate and general inorganic chemical violations make up 92.1% of all MR violations with arsenic making up 7.9%. Although all MR violations are from a failure to conduct regular contaminant monitoring, 99.4%, or repeat/confirmation monitoring, 0.6%, this is not evidence of testing and not reporting. The distinction between not testing and not reporting is subtle compared to testing and not reporting. In order to distinguish testing and not reporting from not testing and not reporting the regulator would have to conduct an investigation of the CWS records, which would have to show the tests were conducted, or the lab where the CWS conducted the test, neither of which may be possible given resource constraints at the regulator.

The violation data does not contain every issued violation to due state regulator underreporting to SDWIS, however the violations that are in the system are very accurate. A survey of 1800 CWSs in 2000 found that 10% of monitoring and reporting violations were in SDWIS but that 95% of the monitoring and reporting violations in SDWIS were accurately reported (*Data Reliability Analysis of the EPA Safe Drinking Water Information System/Federal Version* (2000)).

The SDWIS data contains enforcement actions the regulator took against a CWS in response to an SDWA violation. Table 4.3 summarizes the share of IOC violations that received no enforcement, formal, informal, or resolving enforcement, and statistics on the time it took to return to compliance. IOC MR violations suggestively receive less enforcement than IOC MCL violations. Though this is expected since IOC MCL violations are more severe than MR violations this is not strongly supported by the table due to the small number of IOC MCL violations in the sample. Of the 16 IOC MCL violations, 100%

received some kind of enforcement whereas 2.5% of IOC MR violations received no enforcement action.

A detailed breakdown of the types of enforcement actions used in response to an MR and MCL IOC violation are listed in table 3. Enforcement is either informal or formal. Formal enforcement is more serious, it includes penalties, compliance agreements, and consent decrees. Penalties are rare among the sample although they are used in 0.4% of IOC MR violations. Consent decrees and compliance agreements can be costly even if they do not include penalties. A consent decree may require a system to install new treatment technology. For an example of how expensive a treatment systems is, an EPA best available technology project to replace existing treatment with an arsenic coagulation/filtration system at a CWS with baseline arsenic levels of 0.029 mg/L, serving 250 people in Pennsylvania in 2011 cost \$191,970, including \$136,744 for equipment, \$21,726 for site engineering, and \$33,500 for installation. In addition it cost an average of \$60 a day in chemical, electricity, and labor costs (**epaarsenicbat**).

Informal enforcement actions are the dominant type of enforcement and include public notification requests, reminders and notifications that a violation has occurred, and technical assistance. Informal enforcement includes more direct assistance to the CWS than formal enforcement. The state is the main regulator in the sample. The federal regulator only intervenes to issue informal enforcement in 2.6% of the IOC MR cases and the intervention is to issue a notice of violation.

[h]

Table 1: Days in a Year with IOC Violation

Contaminant	MR Violations		MCL Violations	
	Mean	SD	Mean	SD
Inorganic chemicals	12.27	60.15	0.78	15.83
Nitrates	15.53	67.32	0.18	8.01
Arsenic	10.69	56.01	0.04	2.59

Notes: Sample restricted to strictly downstream community water systems (minehuc_downstream_of_mine=1 and minehuc_mine=0), years 1985–2005. MR = monitoring and reporting violation; MCL = maximum contaminant level violation. Inorganic chemicals encompasses nitrates and arsenic as sub-contaminants. Outcomes measured as days out of the year in violation. N=6,232 CWS×year observations across 340 unique community water systems.

Table 2: IOC Violations by Violation Code and Regulatory Rule, 1985–2005 (Downstream CWSs)

Panel A: MR violations by violation code

Code	Description	Count	Share (%)
<i>Routine and repeat monitoring failures</i>			
03	Monitoring, Regular	1,260	99.0
04	Monitoring, Check/Repeat/Confirmation	13	1.0
<i>Total</i>		1,273	100.0

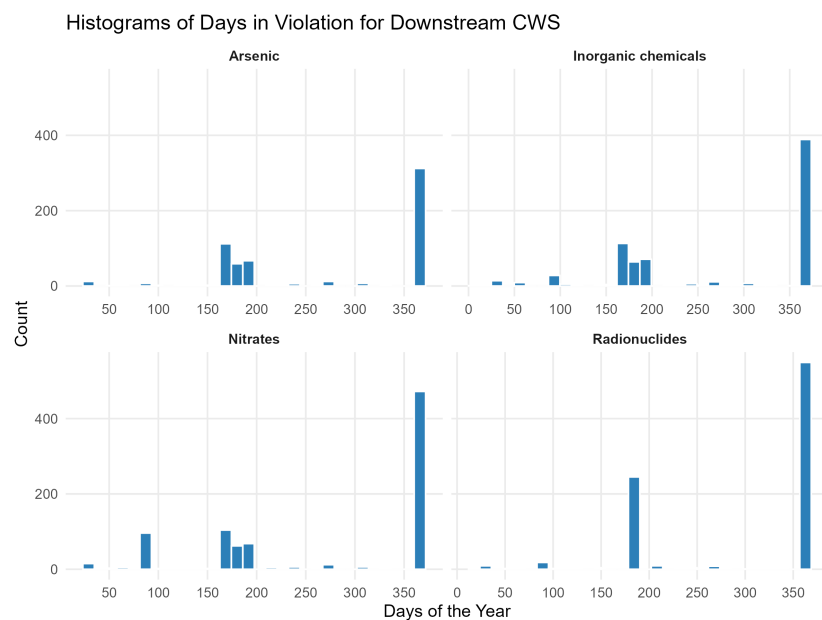
Panel B: MCL violations by violation code

Code	Description	Count	Share (%)
01	Maximum Contaminant Level Violation, Single Sa..	11	68.8
02	Maximum Contaminant Level Violation, Average	5	31.2
<i>Total</i>		16	100.0

Panel C: MR vs. MCL violations by regulatory rule

Rule code	Rule name	MR		MCL	
		Count	Share (%)	Count	Share (%)
333	Inorganic Chemicals	871	68.4	12	75.0
331	Nitrate Rule	300	23.6	2	12.5
332	Arsenic Rule	102	8.0	2	12.5
<i>Total</i>		1,273	100.0	16	100.0

Notes: Panel A: violation codes classify the type of monitoring-and-reporting failure; code 03 = Monitoring, Regular; 04 = Monitoring, Check/Repeat/Confirmation. Panel B: violation codes classify the type of MCL exceedance. Panel C: rule 331 = Nitrate; 332 = Arsenic; 333 = Inorganic Chemicals; MR = monitoring and reporting violation; MCL = maximum contaminant level violation; inorganic chemicals (333) encompasses nitrates and arsenic as sub-contaminants. All panels restricted to IOC rules (nitrate 331, arsenic 332, inorganic chemicals 333). Sample restricted to strictly downstream CWSs (minehuc_downstream_of_mine=1 and minehuc_mine=0), 1985–2005. Source: SDWA_VIOLATIONS_ENFORCEMENT.parquet, SDWA_REF_CODE_VALUES.csv.



Regulator response	IOC MR	IOC MCL
<i>N violations</i>	1,273	16
<i>Enforcement intensity</i>		
No enforcement received (%)	2.5	0.0
Any enforcement received (%)	97.5	100.0
Informal (%)	23.6	31.2
Resolving (%)	66.9	62.5
Formal (%)	6.9	6.2
Federal enforcement (%)	6.6	0.0
Major violation (%)	98.4	0.0
<i>Return to compliance</i>		
Status: Resolved (%)	73.8	68.8
Status: Archived (%)	26.2	31.2
Status: Open (%)	0.0	0.0
Median days to compliance	364	364
Mean days to compliance	34 511	420

Unit of observation: violation (unique VIOLATION_ID). Sample: downstream 2SLS sample (minehuc_downstream_of_mine = 1 and minehuc_mine = 0), 1985–2005 (349 CWSs; states: AL, CA, CO, FL, GA, IL, KS, KY, LA, MD, NC, NJ, NY, OH, OR, PA, SC, TN, UT, VA, WA, WV). IOC rules: nitrate (331), arsenic (332), inorganic chemicals (333). Multiple enforcement actions per violation assigned the most severe (Formal > Resolving > Informal). Days to compliance = CALCULATED_RTC_DATE – NON_COMPL_PER_BEGIN_DATE. Source: SDWA_VIOLATIONS_ENFORCEMENT.parquet.

4.4 CWS Contaminant Measurement Data

In order to measure the effect of upstream mining on actual contaminant concentrations, we construct annual mean contaminant concentrations of several inorganic chemical contaminants in raw and treated CWS water using the second Six-Year Review data (SYR2, sample years 1998-2005).

Starting in 1991, the USEPA began conducting reviews of all National Primary Drinking Water Regulations every six years to inform whether the rules need revisions. The first SYR1 has not been made publicly available. The SYR2 was the second round of the six year review covering 1998 to 2005. The data was collected by the EPA by requesting states to submit the data they were required to collect in order to monitor CWS SDWA compliance. Unrelated to the SYR2 and in keeping compliance with the SDWA, states were required to collect compliance monitoring data but were not required to submit that data to the USEPA alongside the violation reports. The SYR2 data contained contaminant data for 89% of the total population served by a drinking water system. Pennsylvania, Mississippi, Kansas, and Louisiana were not included in the SYR2 partly due to a lack of a national database that received and stored the occurrence of contaminants at CWSs at the time.

We clean the SYR2 data following the procedure of Ravalli et al. (2022). We retain records from CWSs and drop transient and non-transient non-community systems. This removes roughly 23% of raw monitoring records. For each contaminant we compute the national share of records above the maximum laboratory detectable level (MDL) and retain only contaminants detected in more than 10% of records. We replace the observations below the MDL of any contaminant with the $\text{MDL}/\sqrt{2}$. Records reporting a concentration greater than 100 times the contaminant’s MCL are dropped as data-entry errors. Among the IOCs, this keeps arsenic, nitrate, barium, chromium, and selenium, and excludes

Table 3: Specific Enforcement Actions for IOC Violations, 1985–2005 (Downstream 2SLS Sample)

Cat.	Code	Description	IOC MR (%) (<i>N</i> =1,273)	IOC MCL (%) (<i>N</i> =16)
<i>Formal enforcement</i>				
Formal	SFK	State Bilateral Compliance Agreement signed	0.1	2.0
	SFR	State Consent Decree/Judgement	1.3	0.0
	SFL	State Administrative/Compliance Order without penalty issued	1.0	0.0
	SFM	State Administrative Penalty assessed	0.4	0.0
<i>Informal enforcement</i>				
Informal	SIE	State Public Notification requested	15.5	22.0
	SIA	State Violation/Reminder Notice	16.0	18.0
	SFJ	State Formal Notice of Violation issued	16.8	14.0
	SIF	State Public Notification received	8.7	12.0
	SIC	State Technical Assistance Visit	0.4	4.0
	SO8	State Other	0.5	4.0
	SIB	State Compliance Meeting conducted	2.8	0.0
	EFJ	Federal issued Formal Notice of Violation	2.6	0.0
	SO6	State Intentional no-action	0.4	2.0
	—	All other informal actions	2.2	0.0
<i>Resolving (compliance achieved)</i>				
Resolving	SOX	State Compliance achieved	31.1	22.0
	EOX	Federal Compliance achieved	0.3	0.0

Notes: Unit of observation is the enforcement action row in SDWA_VIOLATIONS_ENFORCEMENT.csv. Entries show the share (%) of all enforcement actions within each column group that are of the specified type. MR = monitoring/reporting violation; MCL = maximum contaminant level violation; both restricted to IOC rules (nitrate 331, arsenic 332, inorganic chemicals 333). A violation may generate multiple enforcement actions; rows are enforcement actions, not violations. Enforcement categories follow EPA SDWIS classification: *Formal* = legally binding instruments (administrative orders, consent decrees, penalties, civil/criminal referrals); *Informal* = advisory actions and notices (notices of violation, public notification requests, compliance meetings, technical assistance visits); *Resolving* = closure actions recording that the system returned to compliance. “All other informal actions” aggregates codes each below 1% in every column. Sample: downstream 2SLS sample (minehuc_downstream_of_mine=1 and minehuc_mine=0), 1985–2005 (349 CWSs; states: AL, CA, CO, FL, GA, IL, KS, KY, LA, MD, NC, NJ, NY, OH, OR, PA, SC, TN, UT, VA, WA, WV). Source: SDWA_VIOLATIONS_ENFORCEMENT.parquet, SDWA_REF_CODE_VALUES.csv.

cadmium (1.8% detected), mercury (1.6%), and thallium, whose concentrations are too rarely above detection. All mass-concentration readings are converted to mg/L before averaging.

After cleaning, sample-level records are collapsed to the CWS $\times$ contaminant $\times$ year level. The annual concentration of a particular contaminant is the mean of that year's samples at a particular CWS.

Table 4 includes the summary statistics of the contaminants we include in our analysis. The max value of all contaminant readings are below their MCL. The max arsenic reading of 0.0110 mg/L would exceed the stricter arsenic MCL of 0.01 mg/L that became regulation in 2006 but would not exceed the pre-2006 MCL of 0.05 mg/L. That mean concentrations do not exceed MCL are consistent with the low rates of MCL violations at CWSs. Conditional on testing the water contaminants appear to be within the regulatory limits. However, the majority of MR violations are for not conducting testing on schedule. These summary statistics do not provide information on the contaminant concentrations that would be measured when a test was required by regulation but was not conducted by the CWS.

Table 4: Summary statistics: mean concentration for inorganic chemicals and cumulative upstream coal production

Variable	MCL	Mean	Max	Std. Dev.	<i>N</i>
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0029	0.0110	0.0023	378
Nitrate (mg/L)	10.0 mg/L	0.7520	5.540	0.8482	444
Barium (mg/L)	2.000 mg/L	0.0748	0.7071	0.0671	369
Chromium (mg/L)	0.100 mg/L	0.0059	0.0354	0.0050	374
Selenium (mg/L)	0.050 mg/L	0.0048	0.0354	0.0051	367
Cumul. upstream coal prod. (10M ST)	—	0.4395	14.179	1.415	519

Sample period 1998–2005 (robustness). Statistics computed on the sample used in each chemical's regression in

Table ???. For cumulative upstream coal production, statistics are computed over unique PWSID $\times$ year pairs that appear in at least one regression sample. Sample: CWSs at most one HUC12 downstream of a coal mine

(minehuc_downstream_of_mine = 1, minehuc_mine = 0).

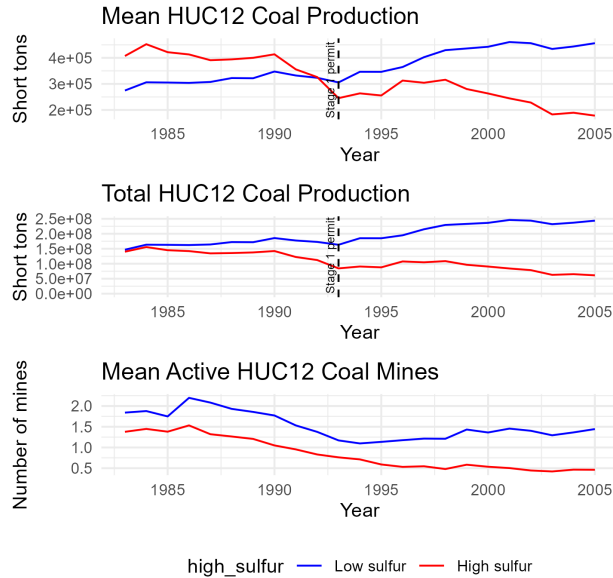
5 The impact of the Acid Rain Program (ARP) on CWS exposure to coal production

We use descriptive and summary data to show that the reduction in high sulfur coal production induced by the ARP also reduced the size of the population reliant on a CWS exposed to coal production. These results do not reflect the actual population served by CWSs affected by coal mines during 1985 to 2005 because we rely on the population served by CWSs using the SDWIS snapshot from the third quarter of 2024.

We find that between 1983 and 2005, tons of coal mined from low sulfur HUC12's increased, while production from high sulfur coal decreased. We consider a HUC12 watershed and all the mines within that watershed to have high sulfur coal if sulfur is greater than or equal to 2%⁴. Figure 5 plots the average trend in HUC12 level coal production by high and low sulfur coal HUC12's from 1983 to 2005. The HUC12 watersheds where coal increased had fewer CWS water intakes downstream than the places where coal production fell. Map 5 plots changes in HUC12 coal production for all HUC12's with at least one active coal mine between 1985 and 2005. The red circles represent proportionate increases in total short tons of coal produced and the blue circles represent proportionate decreases. Coal increased and decreased in both the western and eastern coal fields. Coal production mostly decreased in Pennsylvania and mostly increased in West Virginia. Other state's coal production saw both increases and decreases.

The increases in coal production largely did not occur upstream of CWS water intakes. Map 5 plots coal mining HUC12's upstream of CWS water intakes with circles proportionate to the size of the population that obtains their

⁴Following the USGS literature low sulfur coal is 1 percent or less sulfur by weight, medium sulfur coal is 1 to 3 percent sulfur by weight, and high sulfur coal is greater than 3 percent sulfur by weight Schweinfurth (2009).



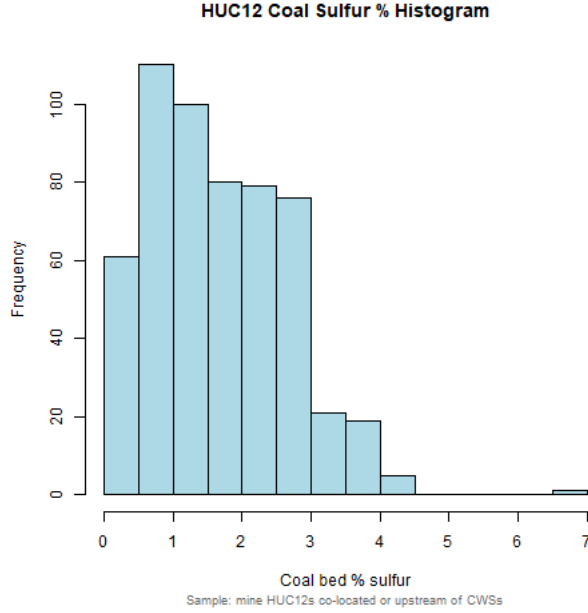
Sample: mine HUC12s co-located or upstream of CWSs

drinking water from a directly downstream HUC12 CWS. HUC12's where coal production fell are plotted blue and where coal production increased are plotted red. HUC12's with no CWS water intake directly downstream are not plotted. The size of the population downstream of HUC12's where coal production fell was larger than the population downstream of HUC12's where coal production increased.

6 The impact of coal mining on CWS SDWA violations

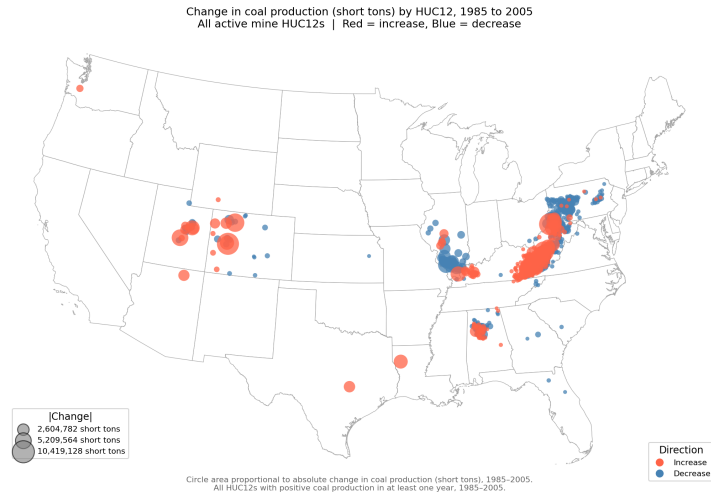
6.1 Empirical Framework

To investigate whether CWSs stay in SDWA compliance with upstream coal mining we regress SDWA violations on the number of active coal mines upstream of the CWS water intake. We instrument for coal production occurring in the



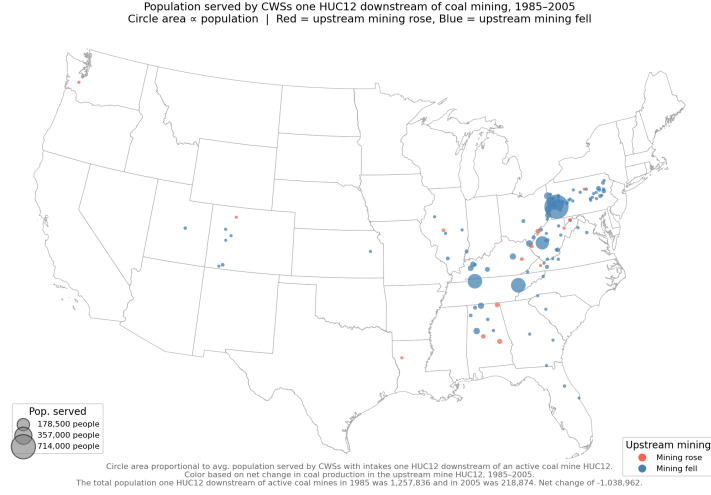
HUC12 directly upstream of a CWS with an instrumental variable constructed by interacting a dummy for the time period after 1995, the compliance date for stage 1 of the ARP, with the percent of sulfur by weight in the coal deposits within the HUC12 directly upstream of the CWS intake. Coal demand for high sulfur coal fell after 1995 as coal fired power plants switched to low sulfur coal to meet the emission standards of the ARP. Production in low sulfur coal areas increased in existing low sulfur coal mines. The instrument identifies the variation in coal production that results from the sudden change in preferences for coal sulfur content that occurred after stage 1 of the ARP.

The instrument is used to address the endogeneity in the location of coal mines with respect to the water quality and behavior of CWSs. In particular we cannot control for all factors that influence both where a mine sites and how capable a CWS is at managing violations which lead to omitted variable bias in an OLS estimate of the impact of coal mines on CWS drinking water qual-



ity violations. The locations of CWS water intakes and coal mines are located non-randomly with respect to each other. For example, coal mines site where they are able to pass environmental reviews, and a CWS downstream of a coal mine will increase the chances of failing an environmental review. A CWS with political power may be able to influence regulators to prevent polluting industries from siting upstream, and those systems are also more capable of avoiding violations, leading to a positive correlation between CWS SDWA violations and upstream coal mining. This bias works in the same direction we expect the true effect of coal mining on CWS SDWA violations to be, making a specification error from a naive OLS difficult to identify.

We estimate the effect of upstream coal mining on CWS drinking water violations using an instrumental variable two stage least squares regression. The first stage is



$$\begin{aligned} \text{NumMines}_{ct} = & \alpha_0 + \alpha_1 \text{NumIntakes}_{ct} + \alpha_2 \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \\ & + Y_t + C_c + \nu_{ct} \end{aligned} \quad (4)$$

Where the Number of Mines that are one HUC12 upstream of CWS c in year t is NumMines_{ct} . The exogenous covariates are the number of CWS intake facilities, NumIntakes_{ct} , year fixed effects, Y_t , and CWS fixed effects, C_c . The instrumental variable is the interaction between the average sulfur as a percentage of weight in the HUC12's directly upstream of all CWS c 's water intakes, $\text{UpstreamCoalSulfur}\%_c$, and a dummy variable equal to 1 for the years after the deadline to implement stage I of the ARP, 1995, and 0 other times, Post95_t . During the pre-1995 period all observations of $\text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t$ are 0 and after 1995 the observations are the percent of sulfur in upstream coal. Standard errors are clustered at the CWS level.

The second stage of the regression is

$$\begin{aligned} \text{Days in Violation}_{vct} = & \beta_0 + \beta_1 \text{NumIntakes}_{ct} + \gamma \widehat{\text{NumMines}}_{ct} \\ & + Y_t + C_c + \epsilon_{ct} \end{aligned} \quad (5)$$

In the second stage, the dependent variable is the days out of a year CWS c experienced violation v in year t , $\text{Days in Violation}_{vct}$.

The endogeneity of upstream sulfur is controlled for with CWS fixed effects. Sulfur is endogenous to water quality. Higher sulfur levels are associated with more sulfide minerals in the rock that oxidize and cause AMD. High sulfide rocks in the geology of a watershed naturally oxidize and leach IOCs into the water, raising the baseline contaminant level, and those rocks are more likely to contain coal seams that attract coal mining. The time invariant ability of a CWSs with greater political power and resources will be able to site their water intakes in areas with cleaner baseline ambient water quality is controlled for with CWS fixed effects.

6.2 Instrument validity

To obtain consistent estimates of β_1 it is necessary for $E[\text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \epsilon_{ct} | Y_t, C_c] = 0$.

$\text{UpstreamCoalSulfur}\% \times \text{Post95}$ is independent of the omitted variables that determine drinking water quality violations. The ARP and the timing of the first phase were both exogenous to CWS drinking water quality and CWS characteristics. The ARP was a national clean air policy that was not drafted or enacted to address to address issues relating to water quality. The effects of the ARP on coal production were determined by national energy markets and not local CWSs or local CWS water quality. The timing of the ARP did not coincide with major drinking water regulations. Local CWS characteristics regarding wa-

ter quality or local power over source protection did not affect the scope of the ARP, nor how stringent the ARP law was, nor the degree to which low sulfur coal should be part of the abatement technology, nor the date of phase I of the ARP. Knowing about sulfur as a percent of coal upstream of CWS after 1995 tells us nothing about how much political power a local PWS has about mine siting. Thus, $Cov(\text{political power of CWS}_c, \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t) = 0$. Assuming that the relationship between CWS characteristics important for water quality and the upstream sulfur content did not change after 1995, then $f(\text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t, \text{CWS characteristic}) = f(\text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t) f(\text{CWS characteristic})$. The instrument would violate independence if the unobserved CWS characteristics that affect drinking water quality violations changed after 1995 according to the amount of upstream coal sulfur percentage.

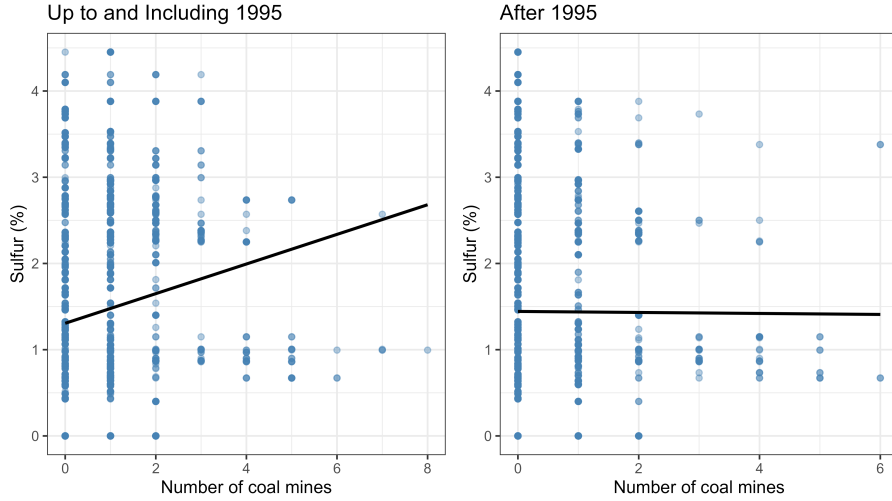
The instrument only affects CWSs through its effect on the number of coal mines. A threat to this would be if something about CWS water quality or SDWA regulation changed after 1995 and had a differential impact on CWSs downstream of high sulfur coal before and after 1995 compared to CWSs downstream of low sulfur coal before and after 1995. Changes in SDWA rules that affected all CWSs the same way are controlled for with time fixed effects. IOC MCL's did not change over this time period, although the arsenic rule was updated in 2001 it did not come into force until 2006. The SDWA amendments that were passed in 1996 increased the use of consumer confidence reports as an enforcement tool when a CWS violated a drinking water regulation. CWSs downstream of high sulfur coal will experience less upstream coal mining after 1995 compared to before 1995 and therefore less upstream water contamination, whereas CWSs downstream of low sulfur coal will experience more upstream coal mining after 1995 compared to before 1995. CWSs downstream of high sulfur coal may experience less SDWA violations from upstream pollution compared to

before 1995 meaning the burden of a consumer confidence report is smaller than it would have been before 1995. Whereas CWSs downstream of low sulfur coal will experience more upstream pollution after 1995 compared to before meaning the burden of the consumer confidence report is higher than it would have been before 1995. This difference is likely small since all CWSs in the sample are downstream of coal mining and will have a more similar pollution exposure and therefore burden from consumer confidence reports compared to the national average CWS.

We can check this by look at relationships between $ARP_{xpost95}$ and cws characteristics not associated with number of upstream mines. We also look at the relationship between violations and $ARP_{xpost95}$ where number of upstream mines is unrelated to $ARP_{xpost95}$

The instrument is predictive of coal production. Figure 6.2 is a scatter plot of the number of coal mines and the coal sulfur percent in the HUC12's upstream of CWSs before and after 1995 and plotted linear lines of best fit. The change in production away from high sulfur coal production after the ARP is strong. The number of coal mines in a HUC12 was increasing with the percent of sulfur as weight of coal in the HUC12 before 1995. After stage 1 of the ARP the number of coal mines in a HUC12 did not strongly increase with the amount of HUC12 coal sulfur content. The time contingent switch in production preferences for coal based on sulfur content supports the structure of the variable $UpstreamCoalSulfur\%_c \cdot Post95_t$. The coefficient of $UpstreamCoalSulfur\%_c \cdot Post95_t$ is partly identified from the relative differences in the differences in the mean number of coal mines of a given sulfur level before and after 1995. The difference in mean coal mines given a high sulfur levels is large before and after 1995 following the fall in production of coal in high sulfur coal beds.

HUC12 sulfur (%) vs. number of coal mines

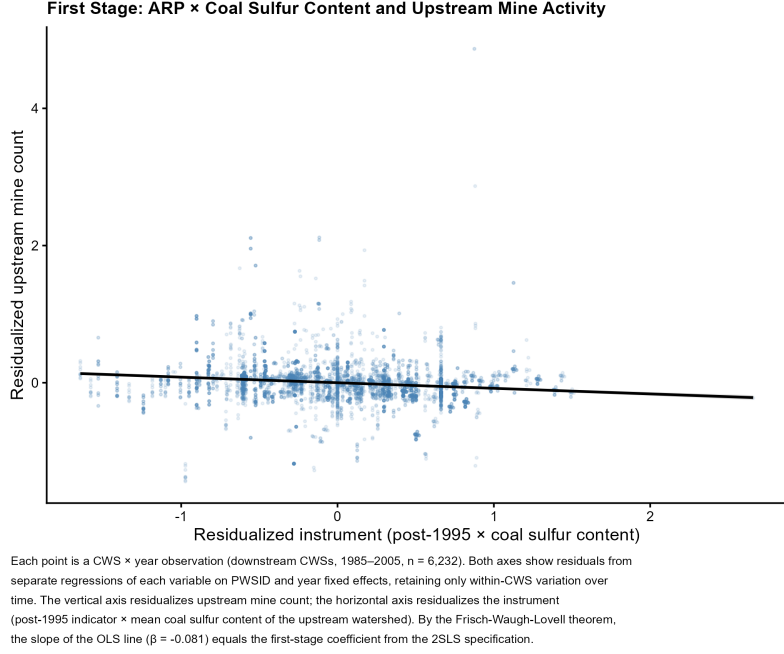


Sample: mine HUC12s (D1) upstream of downstream-only 2SLS CWS intakes, ≥ 1 active mine year 1985-2005.

The strength of the instrument is still evident after the controlling for co-variates. Figure 6.2 is a scatter plot of the residualized number of upstream coal mines and the residualized instrumental variable. There is a negative relationship between the residualized number of upstream coal mines and $UpstreamCoalSulfur\%_c \cdot Post95_t$.

6.2.1 Assessing monotonicity

In order to recover the LATE estimate of the instrument we need the instrument to exhibit monotonicity and for there to be no defiers. A defier is a place that experienced more mining after 1995 because it had higher sulfur. Defiance would require that HUC12 production with high-sulfur would have increased production after ARP but would reduce its production post ARP if it had low-sulfur. A never-taker is a high-HUC12 mine that maintains production because its customer is an electricity generator that installs scrubbers. As long as scrubber adoption is not positively correlated with sulfur in a way that causes production to increase post-ARP in high-sulfur areas, this is not a violation of



monotonicity, and only reduces first-stage power. To be a defier with scrubbers requires a power plant to charge less for power to customers after installing scrubbers, it is not profitable to do so because scrubbers are expensive to install and increase operating costs the profit maximizing response is to raise electricity costs which result in a fall in demand for power and for coal.

If some high-sulfur regions had comparative advantage in low-cost production that offset the ARP cost increase, for example, very low labor costs or transportation advantages, they might expand production even as ARP made their coal relatively more expensive.

Monotonicity follows Angrist and Imbens (1994). Let Ψ be the support of the instrument Z_i , and for each $z \in \Psi$ define a random variable $D_i(z)$. The monotonicity assumption requires that for all $z, w \in \Psi$, either $D_i(z) \leq D_i(w)$ for all i , or $D_i(z) \geq D_i(w)$ for all i . This condition applies pairwise to elements of Ψ and imposes no restriction on the ordering of z and w .

In this setting, the instrument is $Z_i = \text{post95}_t \times \text{sulfur}_h$, which equals zero before 1995 and sulfur_h after. For a given sulfur level w , monotonicity requires either $\text{num_mines}_i(0) \leq \text{num_mines}_i(w)$ for all i , or $\text{num_mines}_i(0) \geq \text{num_mines}_i(w)$ for all i . This permits the direction of the response to vary with sulfur content. For example, at $\text{sulfur} = 1$ (low sulfur), mines may expand after 1995, so monotonicity requires $\text{num_mines}_i(0) \leq \text{num_mines}_i(1)$ for all i . At $\text{sulfur} = 4$ (high sulfur), mines likely contract after 1995, so monotonicity requires $\text{num_mines}_i(0) \geq \text{num_mines}_i(4)$ for all i . These two conditions are compatible: monotonicity is satisfied so long as each pairwise comparison is uniform across units, even when low- and high-sulfur areas respond in opposite directions.

6.3 Results

Table 11 reports the results of equation 5 estimating the impact of the number of active upstream coal mines on the number days out of a year a CWS spends in either an MCL or MR violation for nitrates, arsenic, and inorganic chemical using OLS, reduced form, and two stage least square regressions. An additional active coal mine upstream increases SDWA violations by roughly three days out of the year when we estimate OLS regressions of nitrates, arsenic, and inorganic chemicals on the number of coal mines active upstream coal mines during the year as the main explanatory variable.

Coal production fell most acutely in high sulfur areas after 1995, and a positive relationship between coal production and SDWA violations predicts that coal related violations would fall with increasing upstream sulfur after 1995 relative to before 1995. The reduced form effect of the instrument measures how much more (or less) a one-percentage-point increase in upstream sulfur affects violations after ARP Phase I relative to before. An additional percent

of coal sulfur upstream leads to statistically significantly fewer days a CWS experiences a nitrate, arsenic, or inorganic chemical violation of any kind after 1995 compared to before 1995. This suggests that in areas where coal mining is more likely to close, the number of days in violation falls.

Instrumenting for the number of upstream mines with the upstream sulfur interacted with a post 1995 dummy finds the effect of an additional active mine upstream on nitrates and arsenic violations leads to an additional 19.9 and 14.6 days out of the year with an SDWA violation respectively. The instrument has a relatively strong first stage with an F statistic of 27.5. The OLS results underestimate the effect of coal mines on violations. In general, the OLS impact of coal mines is five times smaller than the two stage least squares regression coefficient, suggesting that the omitted variable bias is negative. A negative omitted variable bias is consistent CWS's that are able to influence the siting of upstream mines being better able at avoiding violations.

The entire increase in violations from upstream mining is driven by MR violations rather than MCL. The regressions presented in table 12 estimates equation 5 on MR arsenic, nitrate, and IOC violations. An additional mine upstream of a CWS leads to 19.9, 14.5, and 14 more days of monitoring and reporting violations for nitrates, arsenic, and general IOC contaminants respectively, comprising the entire effect observed when the dependent variable combines MR violations with MCL violations. IOC violations are statistically significant at the 10% level when violations are limited to MR violations and statistically insignificant when the violations refer to the share of the year spent in either MCL or MR violations.

The impact of coal mining on downstream CWS MCL violations for nitrates, arsenic, and a general category of inorganic chemicals is presented in table 13. The two stage least squares regression of the effect of the number of coal mines

upstream leads to a statistically and economically insignificant changes in the amount of time a CWS spends in an arsenic and nitrate violation. The effect is less than an additional day out of the year. An additional coal mine upstream is associated with two fewer days of maximum contaminant level violations for inorganic chemicals; however, the effect is not statistically significant at any level.

Despite all the significant effects of coal mining being concentrated in monitoring and reporting violations and not in maximum contaminant level violations, we cannot conclude that coal mining does not impact actual contaminant levels. A maximum contaminant level violation is only issued by the regulator when they receive the results of a test a CWS conducted. Consistently failing to conduct tests or report the results of tests and incurring an MR violation prevents an MCL from being issued even if a pollutant legitimately exceeds the MCL. The SDWA ECHO violation dataset is for that reason unable to measure drinking water contamination when systems do not monitor and report tests.

Upstream coal mining increases MR violations without increasing mining does not increase MCL violations. In the framework of the theoretical model SDWA violations are a form of CWS non-compliance with environmental regulation. The theoretical model predicts that non-compliance can increase when the benefits of non-compliance rise and/or the penalty for non-compliance falls. We test whether benefits rise or penalties fall with upstream coal mining.

Table 5: Effect of coal mines on IOC violations (any violation category, CWSs at most one HUC12 down-stream)

	Nitrates			Arsenic			Inorganic chemicals			Radionuclides		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
Upstream coal mines (sum)	3.319*		19.85**	3.501**		14.60*	3.476**		11.79	2.873		-2.390
	(1.810)		(9.491)	(1.701)		(7.953)	(1.751)		(8.697)	(2.295)		(10.82)
post95 × Upstream sulfur %		-4.772**			-3.511*			-2.834			0.5745	
		(2.199)			(1.937)			(2.121)			(2.572)	
F-test (1st stage, clustered), Upstream coal mines (sum)			27.52			27.52			27.52			27.52
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
CWS fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is days out of the year in violation.

Instrument is post95 interacted with sum of coal sulfur content across upstream HUC12s (post95 × sulfur_unified_sum). All regressions include CWS and year fixed effects. Standard errors clustered at CWS level.
Sample period 1985–2005.

Table 6: Effect of coal mines on IOC violations (MR violations only, CWSs at most one HUC12 down-stream)

	Nitrates (MR)			Arsenic (MR)			Inorganic chemicals (MR)			Radionuclides (MR)		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
Upstream coal mines (sum)	3.302*		19.85**	3.511**		14.48*	3.672**		13.96*	2.656		-0.5236
	(1.810)		(9.492)	(1.699)		(7.946)	(1.737)		(8.288)	(2.279)		(10.56)
post95 × Upstream sulfur %		-4.773**			-3.481*			-3.356*			0.1259	
		(2.199)			(1.936)			(2.012)			(2.533)	
F-test (1st stage, clustered), Upstream coal mines (sum)			27.52			27.52			27.52			27.52
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
CWS fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is days out of the year in violation.

Instrument is post95 interacted with sum of coal sulfur content across upstream HUC12s (post95 × sulfur_unified_sum). All regressions include CWS and year fixed effects. Standard errors clustered at CWS level.
Sample period 1985–2005.

Table 7: Effect of coal mines on IOC violations (MCL violations only, CWSs at most one HUC12 down-stream)

	Nitrates (MCL)			Arsenic (MCL)			Inorganic chemicals (MCL)			Radionuclides (MCL)		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
Upstream coal mines (sum)	0.0163		-0.0059	-0.0101		0.1236	-0.1238		-1.951	0.2167		-1.866
	(0.0244)		(0.0063)	(0.0399)		(0.1120)	(0.3553)		(1.910)	(0.2190)		(1.899)
post95 × Upstream sulfur %		0.0014			-0.0297			0.4690			0.4486	
		(0.0015)			(0.0263)			(0.4473)			(0.4480)	
F-test (1st stage, clustered), Upstream coal mines (sum)			27.52			27.52			27.52			27.52
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
CWS fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is days out of the year in violation.

Instrument is post95 interacted with sum of coal sulfur content across upstream HUC12s (post95 × sulfur_unified_sum). All regressions include CWS and year fixed effects. Standard errors clustered at CWS level.
Sample period 1985–2005.

7 The impact of coal mining on downstream CWS contamination

7.1 Empirical Framework

We test the theoretical prediction that non-compliance can increase when upstream mining increases costs of compliance by raising the level of pollution and the effort that needs to be expended to stay in compliance. As costs of compliance rise the cost savings and therefore benefits of non-compliance rise too. We estimate the impact of coal mining on the mean concentration of contaminants within downstream CWSs using the equation

$$Contam\ Concen_{cht} = \beta \cdot \sum_{\tau=1985}^t \text{upstream coal prod}_{c\tau} + \rho_c + HUC02_h \cdot \tau_t + \epsilon_{cht} \quad (6)$$

Where $Contam\ Concen_{cht}$ is the mean concentration of a contaminant measured at CWS c in HUC02 h in year t using SYR2 data from 1997 to 2005. $\sum_{\tau=1985}^t \text{upstream coal prod}_{c\tau}$ is the cumulative tons of coal mined in the HUC12 directly upstream of CWS c from 1985 until year t . β is the effect of an additional 10,000 tons of coal mined upstream of CWS c . CWS fixed effects are included as ρ_c , and to control for annual average pollution levels within the HUC02 watershed, we include HUC02 interacted with time fixed effects $HUC02_h \cdot \tau_t$. Standard errors are clustered at the CWS level.

Equation 6 is a continuous differences in differences with no pure control observations. The parallel trends assumption requires that the pollution concentrations at CWSs that experience a large decline in coal production would trended the same as pollution concentrations at CWSs with a small decline.

Equation 6 provides an unbiased estimate of β if cumulative upstream coal

production is orthogonal to the error term, conditional on the fixed effects:

$$E[\sum_{\tau=1985}^t \text{upstream coal prod}_{c\tau} \epsilon_{cht} | \rho_c HUC02_h \cdot \tau_t] = 0$$

This would be violated if the timing of contaminant concentration tests were affected by upstream coal production. This would be the case if readings were done during periods of low upstream production to avoid detecting contaminants or were done more often when there was a large amount of production leading to large contamination and requiring taking more readings for confirmation. CWS fixed effects control for CWSs level ability to control contamination or time when to take concentration readings. The EPA established federal schedules for when to conduct 6 year reviews which removes some of the concern of CWSs being able to time readings.

7.2 Results

Table 8: Effect of cumulative upstream coal production on Inorganic Chemicals (6-Year Review, downstream CWSs, Ravalli et al. (2022) cleaning, robustness 1998–2005)

	Mean conc.				
	Arsenic (1)	Nitrate (2)	Barium (3)	Chromium (4)	Selenium (5)
Cumul. upstream coal prod. (10M ST)	0.0023*** (0.0004)	0.0572 (0.0922)	0.0171* (0.0093)	0.0027*** (0.0006)	0.0033** (0.0016)
Within R ²	0.03408	0.00075	0.00078	0.00697	0.00741
Observations	334	432	322	328	319
R ²	0.88183	0.88955	0.59983	0.69125	0.57015
CWS fixed effects	✓	✓	✓	✓	✓
huc02-year fixed effects	✓	✓	✓	✓	✓

Sample period 1998–2005 (robustness). Within each chemical, columns show (1) mean measured concentration and (2) share of annual samples exceeding the MCL, both from the EPA 6-Year Review. Non-detect values replaced by $MDL/\sqrt{2}$ following Ravalli et al. (2022). Explanatory variable is cumulative coal production since 1985 (in 10 million short tons) in the HUC12 one step upstream of the CWS intake. Fixed effects: PWSID and HUC02×year (first two digits of intake HUC12 interacted with year). Sample: CWSs at most one HUC12 downstream of a coal mine (minehuc_downstream_of_mine = 1, minehuc_mine = 0). Standard errors clustered at PWSID level.

Table 8 contains the results of regression 6 on inorganic chemical concentra-

tion tests throughout CWS water processing stages. An additional 10 million short tons of cumulative upstream coal production increase mean concentrations of arsenic, nitrate, barium, chromium, and selenium at CWSs. The effect is significant for all IOC contaminants except for nitrates, which we lack the power to identify at a statistically significant level.

Arsenic increases by 0.0023 mg/L after an additional 10 million short tons of coal is mined upstream. This is economically large, equivalent to almost 100% increase over the mean arsenic concentration of 0.0029 mg/L, and statistically significant. The effect represents 23% of the EPA arsenic Maximum Contaminant Level (MCL) of 0.010 mg/L after 2005 and 4.6% of the MCL before 2006 of 0.050 mg/L.

An additional 10 million short tons of coal increases chromium, selenium, and barium by 0.0027 mg/L, 0.0033 mg/L, 0.0171 mg/L respectively. The effects are statistically significant and economically significant, relative to the mean contaminant levels the effects are increases of 50%, 69%, and 24% respectively.

We lack the power to identify the effect of cumulative coal mining on nitrates. An additional 10 million short tons of coal increases nitrate concentrations by 0.05 mg/L but the effect is not economically or statistically significant, the effect is a 5% increase in concentration levels relative to the mean contaminant concentration 0.75 mg/L.

Despite the economic and statistically significant result that upstream coal mining leads to increases in IOC contaminant concentrations at CWSs there are no observations that exceed the the MCL of any inorganic chemicals, see table 4. This does not confirm that water contamination never exceeded the MCL. Most MR violations were issued for failure to test water quality, the SYR2 is necessarily the results from water tests and do not reflect the tests not undertaken and subsequently punished with an MR violation.

Mining increases mean inorganic chemical contamination (by 20-100% of mean concentrations). Mining raises CWS costs to completely remove contaminants. CWSs downstream of mines face costlier SDWA compliance.

8 Regulator Responses to CWS's impacted by coal mining

The theoretical model comparative statics predict that CWSs will violate SDWA more often if the penalty for violating falls. We test whether penalties at CWSs fall when upstream coal mining increases.

8.1 Empirical Strategy

We measure enforcement severity on a relative scale where formal enforcement is more severe than informal enforcement. We estimate the effect of upstream coal mining on the likelihood of formal and informal enforcement with an instrumental variable two stage least squares regression. In addition we estimate the effect of upstream coal mining on the likelihood of a sanitary visit during the year by the regulator to the CWS. The number of upstream coal mines is instrumented with $\text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t$ as in equation 4.

The first stage is

$$\begin{aligned} \text{NumMines}_{ct} = & \alpha_0 + \alpha_1 \text{NumIntakes}_{ct} + \alpha_2 \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \\ & + Y_t + C_c + \nu_{ct} \end{aligned} \tag{7}$$

And the second stage is

$$\begin{aligned} \text{Regulatory Action}_{ct} = & \beta_0 + \beta_1 \text{NumIntakes}_{ct} + \gamma \widehat{\text{NumMines}}_{ct} \\ & + Y_t + C_c + \epsilon_{ct} \end{aligned} \quad (8)$$

The Number of Mines that are one HUC12 upstream of a CWS c water intake in year t , NumMines_{ct} , are predicted by a function of the number of CWS intake facilities, NumIntakes_{ct} , the average coal sulfur percentage in upstream HUC12's where coal sulfur can be calculated, $\text{UpstreamCoalSulfur}\%_c$, interacted with a dummy indicator for the years after 1995, stage I of the ARP, Post95_t , year fixed effects, Y_t , CWS fixed effects, C_c , and an error term, ν_{ct} . Standard errors are clustered at the CWS level.

Any effect the instrument has on enforcement is driven through the causal chain from the effect of coal mining on water pollution to the regulator's response to that pollution. The exclusion restriction fails when the instrument affects regulator behavior through another path outside of its affect on pollution through changing coal mining.

The exclusion restriction would be threatened if the ARP Phase I directly changed state EPA enforcement practices in a way that affected CWSs differently according to the percent of sulfur by weight in upstream coal. However, states were not primarily responsible for enforcing the ARP, it was a federally regulated program which by itself would not have affected state primacy enforcement of the SDWA.

Regulators are more likely responding to pollution rather than upstream coal mining levels since coal mines production is regulated by state and federal revenue authorities and mine safety is regulated the federal MSHA. It is possible that states environmental regulators are aware of coal mining pollution through their primacy to enforce point source pollution in the Clean Water Act. Coal

mines are point source polluters.

The CWS fixed effects control for the effect that upstream sulfur is associated with a higher baseline of AMD which could increase pollution and therefore enforcement action outside of the effect of the instrument on coal mines.

Any national regulatory changes to the SDWA that affected all CWSs are controlled for with time fixed effects. The threat to the exogeneity of the instrument would occur if regulatory enforcement at low sulfur regions before and after 1995 changed differently to the change in regulatory enforcement at high sulfur regions before and after 1995.

8.2 Results

Upstream coal mining leads to a reduction in the costliest enforcement action. Table 9 presents the coefficients from estimating equation 8 on the likelihood of formal and informal enforcement. An additional upstream coal mine leads to a 5.6 percentage point fall in the likelihood of a CWS receiving formal enforcement action. We lack the power to identify a statistically significant effect of the number of upstream coal mines on informal enforcement rates, although the effect is economically small at -0.0056 percentage points.

Table ?? presents the coefficients from estimating equation 8 on the likelihood of any sanitary visit to a CWS caused by an increase in upstream coal mining. Coal mining leads to a 14 percentage point increase in sanitary visits by the regulator to the CWS. This is statistically significant at the 1% level. Sanitary visits are a remedial intervention the regulator can use to assess the severity of a violation. Sanitary visits are less punitive than formal enforcement action.

Coal mining reduces the costliest type of enforcement action on CWSs. Upstream coal mining increases the regulators use of less costly enforcement action

in the form of sanitary visits. These results are consistent with the theoretical model that predicts that the increase in SDWA non-compliance is also driven by CWSs facing lower penalties for violations.

Table 9: Effect of Coal Mining on Enforcement Actions by Type (D1 Downstream Sample)

	Any informal enf			Any formal enf		
	OLS	RF	2SLS	OLS	RF	2SLS
Upstream coal mines (sum)	0.0084 (0.0079)		-0.0056 (0.0400)	-0.0169*** (0.0045)		-0.0565*** (0.0204)
post95 $\times$ Upstream sulfur %		0.0014 (0.0102)			0.0145*** (0.0052)	
F-test (1st stage, clustered), Upstream coal mines (sum)			27.94			27.94
Observations	6,225	6,225	6,225	6,225	6,225	6,225
CWS fixed effects	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓

D1 downstream sample (minehuc_downstream_of_mine = 1, minehuc_mine = 0). Cols 1-3: informal enforcement action (19.0% of panel). Cols 4-6: formal enforcement action (3.5% of panel). Treatment: num_coal_mines_upstream_sum. Instrument: post95 x sulfur_unified_mean. SEs clustered at CWS level.

Table 10: Effect of Coal Mining on Sanitary Survey Probability (D1 Downstream Sample, LPM)

	Any sanitary visit		
	OLS	RF	2SLS
Upstream coal mines (sum)	0.0106 (0.0067)		0.1406*** (0.0487)
post95 $\times$ Upstream sulfur %		-0.0360*** (0.0103)	
F-test (1st stage, clustered), Upstream coal mines (sum)			27.94
Observations	6,225	6,225	6,225
CWS fixed effects	✓	✓	✓
year fixed effects	✓	✓	✓

D1 downstream sample (minehuc_downstream_of_mine = 1, minehuc_mine = 0). Outcome: any sanitary survey (SNSV) in CWS-year. N = 6232 CWS-years. Treatment: num_coal_mines_upstream_sum. Instrument: post95 x sulfur_unified_mean. SEs clustered at CWS level.

9 Conclusion

Upstream coal mining induces lower CWS SDWA compliance. An increase in mining leads to increases in inorganic chemical water contamination at CWSs. The increase in inorganic chemical water contamination raises the cost off completely removing contaminants but since we do not detect mean contaminant levels increasing above maximum contaminant levels we cannot conclude that upstream coal mining is increasing pollution to the extent that CWSs need to install improved abatement technology to meet SDWA MCL compliance. We do identify that upstream coal mining leads to falls in formal enforcement action, which is the costliest form of enforcement, and an increase in sanitary visits, which is a non-punitive and remedial enforcement action. The increase in CWS SDWA non-compliance with an increase in upstream coal mining is consistent with lower punishment for violations and higher costs of compliance.

10 Appendices

10.1 Alternative Specification of Number of Upstream Coal Mines

Table 11: Effect of coal mines on mining violations (any violation category, CWSs at most one HUC12 down-stream)

	Nitrates			Arsenic			Inorganic chemicals			Radionuclides		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
num_coal_mines_upstream_mean	6.088 (4.017)		60.07** (30.38)	6.652 (4.129)		45.39* (24.82)	6.758 (4.154)		37.25 (26.61)	8.358* (4.845)		-3.520 (33.68)
post95 × sulfur_unified_mean		-4.937** (2.230)			-3.730* (1.981)			-3.062 (2.167)			0.2893 (2.751)	
F-test (1st stage, clustered), num_coal_mines_upstream_mean			13.68			13.68			13.68			13.68
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
R ²	0.15763	0.15860	0.15860	0.17190	0.17204	0.17204	0.18745	0.18714	0.18714	0.24894	0.24759	0.24759
Within R ²	0.00089	0.00205	0.00205	0.00160	0.00176	0.00176	0.00137	0.00099	0.00099	0.00180	7.72×10^{-6}	7.72×10^{-6}
PWSID fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is days out of the year in violation.

Instrument is post95 interacted with mean coal sulfur content of the intake watershed (post95 x sulfur unified).

All regressions include PWSID and year fixed effects. Standard errors clustered at PWSID level. Sample period 1985–2005.

Table 12: Effect of coal mines on mining violations (MR violations only, CWSs at most one HUC12 down-stream)

	Nitrates (MR)			Arsenic (MR)			Inorganic chemicals (MR)			Radionuclides (MR)		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
num_coal_mines_upstream_mean	6.072 (4.015)		60.08** (30.39)	6.651 (4.124)		45.01* (24.77)	6.968* (4.193)		43.59* (25.87)	7.784 (4.779)		1.902 (32.93)
post95 × sulfur_unified_mean		-4.938** (2.230)			-3.699* (1.980)			-3.582* (2.068)			-0.1563 (2.714)	
F-test (1st stage, clustered), num_coal_mines_upstream_mean			13.68			13.68			13.68			13.68
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
R ²	0.15531	0.15630	0.15630	0.17235	0.17246	0.17246	0.18175	0.18166	0.18166	0.25068	0.24949	0.24949
Within R ²	0.00089	0.00207	0.00207	0.00160	0.00173	0.00173	0.00153	0.00142	0.00142	0.00159	2.71×10^{-6}	2.71×10^{-6}
PWSID fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is days out of the year in violation.

Instrument is post95 interacted with mean coal sulfur content of the intake watershed (post95 x sulfur unified).

All regressions include PWSID and year fixed effects. Standard errors clustered at PWSID level. Sample period 1985–2005.

Table 13: Effect of coal mines on mining violations (MCL violations only, CWSs at most one HUC12 down-stream)

	Nitrates (MCL)			Arsenic (MCL)			Inorganic chemicals (MCL)			Radiomclides (MCL)		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
num_coal_mines_upstream_mean	0.0155 (0.0332)		-0.0135 (0.0153)	0.0004 (0.0496)		0.3743 (0.3563)	-0.0822 (0.5632)		-5.601 (5.643)	0.5743 (0.5841)		-5.422 (5.626)
post95 × sulfur_unified_mean		0.0011 (0.0012)			-0.0308 (0.0281)			0.4603 (0.4440)			0.4456 (0.4451)	
F-test (1st stage, clustered), num_coal_mines_upstream_mean			13.68			13.68			13.68			13.68
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
R ²	0.37584	0.37584	0.37584	0.05303	0.05398	0.05398	0.12889	0.12916	0.12916	0.19493	0.19539	0.19539
Within R ²	5.6×10^{-7}	1.32×10^{-8}	1.32×10^{-8}	8.3×10^{-7}	4.92×10^{-5}	4.92×10^{-5}	3.8×10^{-6}	0.00032	0.00032	0.00052	0.00109	0.00109
PWSID fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is days out of the year in violation.

Instrument is post95 interacted with mean coal sulfur content of the intake watershed (post95 × sulfur unified).

All regressions include PWSID and year fixed effects. Standard errors clustered at PWSID level. Sample period 1985–2005.

Table 14: Effect of coal mines on IOC violations (any violation category, CWSs at most one HUC12 down-stream)

	Nitrates			Arsenic			Inorganic chemicals		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
Upstream coal mines (sum)	0.0194*** (0.0072)		0.0991*** (0.0372)	0.0167** (0.0065)		0.0751** (0.0306)	0.0175*** (0.0065)		0.0634* (0.0332)
post95 × Upstream sulfur %		-0.0238*** (0.0084)			-0.0180** (0.0073)			-0.0152* (0.0080)	
F-test (1st stage, clustered), Upstream coal mines (sum)			27.52			27.52			27.52
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
CWS fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is an indicator equal to 1 if the CWS

had any violation of that type during the year, 0 otherwise. Instrument is post95 interacted with sum of coal sulfur content across upstream HUC12s (post95 × sulfur_unified_sum). All regressions include CWS and year fixed effects. Standard errors clustered at CWS level. Sample period 1985–2005.

Table 15: Effect of coal mines on IOC violations (MR violations only, CWSs at most one HUC12 down-stream)

	Nitrates (MR)			Arsenic (MR)			Inorganic chemicals (MR)		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
Upstream coal mines (sum)	0.0195*** (0.0072)		0.0993*** (0.0373)	0.0170*** (0.0065)		0.0743** (0.0305)	0.0180*** (0.0066)		0.0707** (0.0323)
post95 × Upstream sulfur %		-0.0239*** (0.0084)			-0.0179** (0.0073)			-0.0170** (0.0077)	
F-test (1st stage, clustered), Upstream coal mines (sum)			27.52			27.52			27.52
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
CWS fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is an indicator equal to 1 if the CWS

had any violation of that type during the year, 0 otherwise. Instrument is post95 interacted with sum of coal sulfur content across upstream HUC12s (post95 × sulfur_unified_sum). All regressions include CWS and year fixed effects. Standard errors clustered at CWS level. Sample period 1985–2005.

Table 16: Effect of coal mines on IOC violations (MCL violations only, CWSs at most one HUC12 down-stream)

	Nitrates (MCL)			Arsenic (MCL)			Inorganic chemicals (MCL)		
	OLS	RF	2SLS	OLS	RF	2SLS	OLS	RF	2SLS
Upstream coal mines (sum)	-0.0002 (0.0002)		-0.0002 (0.0002)	-0.0002 (0.0004)		0.0008 (0.0006)	-4.42×10^{-5} (0.0014)		-0.0078 (0.0069)
post95 $\times$ Upstream sulfur %		4.69×10^{-5} (4.95×10^{-5})			-0.0002 (0.0001)			0.0019 (0.0016)	
F-test (1st stage, clustered), Upstream coal mines (sum)			27.52			27.52			27.52
Observations	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225	6,225
CWS fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓

Columns show OLS, reduced form, and 2SLS estimates. Dependent variable is an indicator equal to 1 if the CWS had any violation of that type during the year, 0 otherwise. Instrument is post95 interacted with sum of coal sulfur content across upstream HUC12s (post95 $\times$ sulfur_unified_sum). All regressions include CWS and year fixed effects. Standard errors clustered at CWS level. Sample period 1985–2005.

Table 17: Concurrent and Following Violation after an MR Violation, by Contaminant Group, 1985–2005

	Mining MR	Non-mining MR
<i>Panel A: Concurrent period — same PWSID, same contaminant, same year</i> ($N = 523$ mining; $N = 741$ non-mining PWSID $\times$ rule $\times$ year cells)		
MR only — no concurrent MCL or TT (%)	99.2	93.3
MCL concurrent with MR (%)	0.8	5.1
TT concurrent with MR, no MCL (%)	0.0	1.6
<i>Panel B: Following year — same PWSID, same contaminant, year + 1</i> (index violations in 1985–2004; $N = 506$ mining; $N = 678$ non-mining)		
No violation recorded (%)	94.1	74.6
MR only — monitoring failure again, no MCL (%)	5.9	20.8
MCL violation (with or without MR) (%)	0.0	3.2
TT only — no MR or MCL (%)	0.0	1.3

Notes: Unit of observation is the PWSID $\times$ rule code $\times$ calendar year cell. Each cell is classified as 1 if any violation of the stated type was recorded in SDWIS for that PWSID, rule, and year; 0 otherwise. Index cells are those with at least one MR violation. Mining-related rule codes: nitrate (331), arsenic (332), inorganic chemicals (333), radionuclides (340). Non-mining rule codes: total coliform (110, 111), surface/groundwater rule (121–123, 140), VOCs (310), SOCs (320). *Panel A* rows are mutually exclusive and sum to 100: the index cell always has MR=1, so concurrent MCL or TT indicates a second type of violation for the same PWSID, contaminant, and year alongside the monitoring failure. *Panel B* rows are mutually exclusive and sum to 100: MCL takes precedence over MR and TT; TT only requires no MCL and no MR in year + 1. “No violation” in Panel B means no SDWIS record for that PWSID and contaminant rule in year + 1; this includes both genuine compliance and cases where the PWSID left the sample. Index violations in 2005 are excluded from Panel B (year + 1 falls outside the sample period). Source: SDWA_VIOLATIONS_ENFORCEMENT.csv.

Table 18: Violation Following an Arsenic, Nitrate, or Inorganic Chemical MR Violation, Downstream CWSs, 1985–2005

Next violation	Mining MR (<i>N</i> =1,273)
<i>No subsequent violation</i>	
No further violation in 1985–2005 (%)	2.2
<i>Monitoring/reporting violation follows</i>	
MR, same contaminant ^a (%)	4.9
MR, same contaminant group ^b (%)	9.8
MR, other contaminant (%)	63.1
<i>Limit exceedance violation follows</i>	
MCL, same contaminant ^a (%)	0.2
MCL, same contaminant group ^b (%)	0.3
MCL, other contaminant (%)	6.0
<i>Other violation type follows</i>	
Treatment technique (TT) (%)	2.1
Other category (%)	11.4
<i>Time to next violation</i>	
Median days to next violation	365.0
Mean days to next violation	674.7

Notes: Unit of observation is the MR violation (unique VIOLATION_ID). Sample: CWS in the coal mining analysis panel whose intake HUC is one step downstream of a mine HUC (downstream-only sample), 1985–2005. Mining-related rules: nitrate (331), arsenic (332), inorganic chemicals (333). Non-mining rules: total coliform (110, 111), surface/groundwater rule (121–123, 140), VOCs (310), SOCs (320). “Next violation” is defined as the highest-severity violation beginning on the next distinct begin date for the same PWSID; severity order is MCL mining > MCL non-mining > MCL other > TT > MR mining > MR non-mining > other. When multiple violations share the next date, the highest-severity is selected; this is conservative for detecting MCL escalation (the MCL is surfaced if present). “Days to next violation” is missing for violations with no subsequent violation. ^aSame contaminant = same SDWIS RULE_CODE (e.g., an arsenic MR followed by an arsenic MCL). ^bSame contaminant group (but different rule code): e.g., a nitrate MR followed by an arsenic MCL; or a total-coliform MR followed by a surface-water rule MCL. Source: SDWA_VIOLATIONS_ENFORCEMENT.csv.

Table 19: SDWA Datasets Available for Regulator Behavior Analysis (Downstream CWSs)

Dataset	Key Variables	Rows (1985–2005)		Usability
		Total file	Matched sample	
SDWA_SITE_VISITS	Visit frequency, reason code, agency type, eval codes	2,394,287	2,159 (271 PWSIDs)	Strong
SDWA_VIOLATIONS_ENFORCEMENT	Enforcement category, originator, days to RTC, violation status	14,665,006	19,854 (269 PWSIDs)	Strong
SDWA_EVENTS_MILESTONES	Milestone codes (DEEM, DONE, SDFF, FICF), reason codes	262,540	409 (208 PWSIDs)	Narrow (LCR only)
SDWA_PN_VIOLATION_ASSOC	Public notice tier, related violation	355,197	24 (10 PWSIDs)	Too sparse
SDWA_PUB_WATER_SYSTEMS	Reduced monitoring status, outstanding performer	—	349 PWSIDs (cross-sectional)	Too sparse

Note: “Matched sample” = PWSIDs in the strictly downstream coal mining CWS sample.

LCR = Lead and Copper Rule. RTC = return to compliance.

10.2 Binary coded SDWA violations

10.3 Regulator action, violation sequencing, SDWA rule enactment dates

References

- Acid Rain Program Results* (Feb. 2026). USEPA. URL: <https://www.epa.gov/acidrain/acid-rain-program-results>.
- Akcil, Ata and Soner Koldas (2006). “Acid Mine Drainage (AMD): causes, treatment and case studies”. In: *Journal of cleaner production* 14.12-13, pp. 1139–1145.
- Allaire, Maura, Haowei Wu, and Upmanu Lall (2018). “National trends in drinking water quality violations”. In: *Proceedings of the National Academy of Sciences* 115.9, pp. 2078–2083.

Table 20: Site Visits in Analysis Sample, 1985–2005 (Downstream CWSs)

	Count	Share (%)
<i>Panel A: Visit reason</i>		
Sanitary survey (SNSV)	1,053	48.8
Technical assistance (TECH)	284	13.2
Site visit (SITE)	194	9.0
Research (RSCH)	120	5.6
Sampling (SMPL)	139	6.4
Investigation (INVG)	98	4.5
Other (OTHR)	107	5.0
Infrastructure inspection (INFI)	59	2.7
Other (ENGR, PRMT, OM, EMRG, etc.)	105	4.9
<i>Total</i>	2,159	100.0
<i>Panel B: Agency conducting visit</i>		
State (ST)	1,232	57.1
State administrative district (SA)	72	3.3
State administrative region (SR)	33	1.5
District (DS)	5	0.2
Other	4	0.2
<i>Total</i>	2,159	100.0
<i>Panel C: Compliance evaluation outcome</i>		
Not evaluated / not applicable (X, Z)	2,096	97.1
No deficiencies (N)	59	2.7
Recommendations made (R)	4	0.2
<i>Total</i>	2,159	100.0
<i>Panel D: Visits per PWSID-year</i>		
Minimum	1	
25th percentile	1	
Median	1	
Mean	1.8	
75th percentile	2	
Maximum	25	

Note: Sample is 271 unique strictly downstream PWSIDs matched to coal mining analysis sample.

Andarge, Tihitina et al. (Sept. 2025). “Quantifying Threshold Manipulation in the Presence of Rounding: The Case of Lead Monitoring in US Drinking Water”. In: *American Economic Review: Insights* 7.3, pp. 285–305. DOI: 10.1257/aeri.20240258. URL: <https://www.aeaweb.org/articles?id=10.1257/aeri.20240258>.

Angrist, Joshua and Guido Imbens (1994). “Estimation and identification of local average treatment effects”. In: *Econometrica*.

Table 21: Enforcement Actions in Analysis Sample, 1985–2005 (Downstream CWSs)

	Count	Share (%)
<i>Panel A: Violation category</i>		
Monitoring & Reporting (MR)	17,231	86.8
Maximum Contaminant Level (MCL)	949	4.8
Treatment Technique (TT)	875	4.4
Other	799	4.0
<i>Total violations</i>	19,854	100.0
<i>Panel B: Enforcement action category</i>		
Informal	12,706	64.6
Resolving	6,178	31.4
Formal	773	3.9
<i>Total enforcement actions</i>	19,657	100.0
<i>Panel C: Enforcement originator</i>		
State	18,929	96.3
Federal	728	3.7
<i>Total enforcement actions</i>	19,657	100.0
<i>Panel D: Violation status</i>		
Resolved	17,200	86.6
Archived	2,648	13.3
Addressed (formal enforcement, not resolved)	2	0.0
Unaddressed	4	0.0
<i>Total violations</i>	19,854	100.0
<i>Panel E: Days to return to compliance (N = 19,848)</i>		
Minimum	−3, 103	
25th percentile	89	
Median	183	
Mean	330	
75th percentile	364	
Maximum	4,691	

Note: Sample is 269 unique strictly downstream PWSIDs with enforcement actions.

Negative values of days to return to compliance reflect data entry inconsistencies in SDWIS.

Aragón, Fernando M and Juan Pablo Rud (2016). “Polluting industries and agricultural productivity: Evidence from mining in Ghana”. In: *The Economic Journal* 126.597, pp. 1980–2011.

Austin, Wes (2020). “Throwing the Baby out With the Ashwater? Coal Combustion Residuals, Water Quality, and Fetal Health”. In: *Self-published job mar-*

Table 22: SDWA Rule Enactment Dates: Original Rules and Subsequent Revisions

Outcome	Original rule	Effective	Revision (cited previously)
<i>Panel A: Rules in effect before 1985 sample start</i>			
Nitrate	1975 NPDWR (40 FR 59570, Dec 24, 1975); MCL = 10 mg/L	Jun 24, 1977	Phase II Rule (56 FR 3526, Jan 30, 1991) reaffirmed same MCL
Arsenic	1975 NPDWR; MCL = 50 ppb (adopted from 1942 Public Health Service standard)	Jun 24, 1977	Arsenic Rule (66 FR 6976, Jan 22, 2001) lowered MCL to 10 ppb; compliance Jan 23, 2006
Inorganic chemicals	1975 NPDWR; MCLs for barium, cadmium, chromium, lead, mercury, selenium, fluoride, and others	Jun 24, 1977	Phase II/V Rules (1991–1992) added contaminants; core framework unchanged
Radionuclides	1976 Interim Radionuclides Rule; covered gross alpha, Ra-226/228, beta/photon activity	Jun 24, 1977	Radionuclides Rule (65 FR 76708, Dec 7, 2000) revised MCLs; compliance Dec 8, 2003
Total coliform	1975 NPDWR; coliform monitoring standard	Jun 24, 1977	Total Coliform Rule (54 FR 27544, Jun 29, 1989) restructured monitoring; compliance Jan 1, 1990
<i>Panel B: Rules enacted during 1985–2005 sample period</i>			
Surface/groundwater rule (SWTR)	Surface Water Treatment Rule (54 FR 27526, Jun 29, 1989); no pre-1989 equivalent	Jul 1, 1993	Enhanced SWTR (2002); LT1ESWTR (2005)
VOCs	Phase I Rule (52 FR 25690, Jul 8, 1987); first MCLs for benzene, TCE, carbon tetrachloride, vinyl chloride, etc.	Jan 1, 1989	New regulation; not a revision
SOCs	Phase II Rule (56 FR 3526, Jan 30, 1991) and Phase V Rule (57 FR 31776, Jul 17, 1992); first MCLs for atrazine, pesticides, etc.	Jan 1, 1993	New regulation; not a revision

Notes: Panel A outcomes (nitrate, arsenic, inorganic chemicals, radionuclides, total coliform) were regulated under the original National Primary Drinking Water Regulations (NPDWR), effective June 24, 1977, and are present throughout the 1985–2005 study period. The dates commonly cited for these contaminants (Phase II 1991, Arsenic Rule 2001, Radionuclides Rule 2000) refer to revisions that refined MCLs or expanded contaminant lists, not to initial regulation. Panel B outcomes (SWTR, VOCs, SOC) represent genuinely new regulatory requirements introduced during the sample period. SMF = Standardized Monitoring Framework, established under the Phase II Rule (1991) with monitoring beginning January 1, 1993; it consolidated VOC, SOC, and inorganic chemical monitoring into a uniform national schedule.

Table 23: NPDWR Changes Relevant to Pre/Post-1995 IV Comparison

Contaminant	MCL	MR Schedule	TT Requirement	Source & Effective Date
Arsenic	0.05 mg/l	Not revised in this rule	None	1975 NIPDWR; not changed
Inorganic Chemicals (Cd, Cr, Hg, Se, Asbestos)	Cd: 0.005; Cr: 0.1; Hg: 0.002; Se: 0.05; Asbestos: 7 MFL	GW: 1 sample/3-yr period. SW: annually. Waiver available (up to 9-yr). Exceed MCL → quarterly; min 2 (GW) or 4 (SW) before returning to base.	None	This rule; effective July 1, 1992 ; monitoring begins Jan 1, 1993
Nitrates	10 mg/l (as N); joint NO ₃ +NO ₂ : 10 mg/l	GW CWS/NTWS: annually; SW: quarterly. TWS: annually. No waiver . >50% MCL triggers quarterly ≥1 yr. New 24-hr confirmation sample if MCL exceeded.	None	Revised; effective July 30, 1993
Nitrite	1 mg/l (as N) — <i>new standard</i>	1 sample in first compliance period (1993–1995). If <50% MCL: State discretion. If ≥50% MCL: quarterly then annual.	None	New; effective July 30, 1993
Radionuclides	Gross α: 15 pCi/l; Ra-226+228: 5 pCi/l; Beta: 4 mrem/yr	Not revised in this rule	None	1976 NIPDWR; changed
SOCs (18 pesticides/PCBs)	0.00005–0.07 mg/l (see Table 3)	4 consecutive quarterly samples/compliance period starting Jan 1, 1993. >3,300 population: reduce to 2 qtly/yr after clean initial period. ≤3,300: 1 sample/period. 3-yr waiver available. Detection at MDL → quarterly.	Acrylamide: 0.05% @ 1 ppm. Epichlorohydrin: 0.01% @ 20 ppm. Annual written certification required.	New contaminants ; effective July 30, 1992 ; monitoring begins Jan 1, 1993
VOCs (10 new + 8 prior)	0.005–10 mg/l (see Table 2)	4 quarterly samples/compliance period starting Jan 1, 1993. No detect → annual. 3 yrs annual no detect: GW may reduce to 1/period. 6-yr waiver available. Detect (>0.0005 mg/l) → quarterly. Prior 8 VOCs harmonized into same 3/6/9-yr cycle.	None	10 new VOCs; effective July 1, 1992 ; 8-yr cycle harmonized Jan 1, 1993
Total Coliform	Presence/absence (zero tolerance)	Not revised in this rule	None	Total Coliform Rule; effective Dec 31, 1992
Surface/Ground Water Rule	Treatment technique	Not revised in this rule	Disinfection, filtration, turbidity limits	SWTR; effective Dec 31, 1992

Notes: GW = groundwater system; SW = surface water system; CWS = community water system; NTWS = non-transient non-community water system; TWS = transient non-community water system; MFL = million fibers per liter; MDL = method detection limit. The 1991 rule (49 CFR Parts 141, 142, 143) substantially expanded administrative monitoring burden beginning January 1, 1993 — before the ARP Phase I instrument (1995). The second compliance period (1996–1998) creates cross-sectional heterogeneity: systems with no detections in 1993–1995 qualified for reduced monitoring; systems with detections were locked into quarterly sampling.

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